4MRNA User Manual

Last updated date: 2025.08.01

Overview

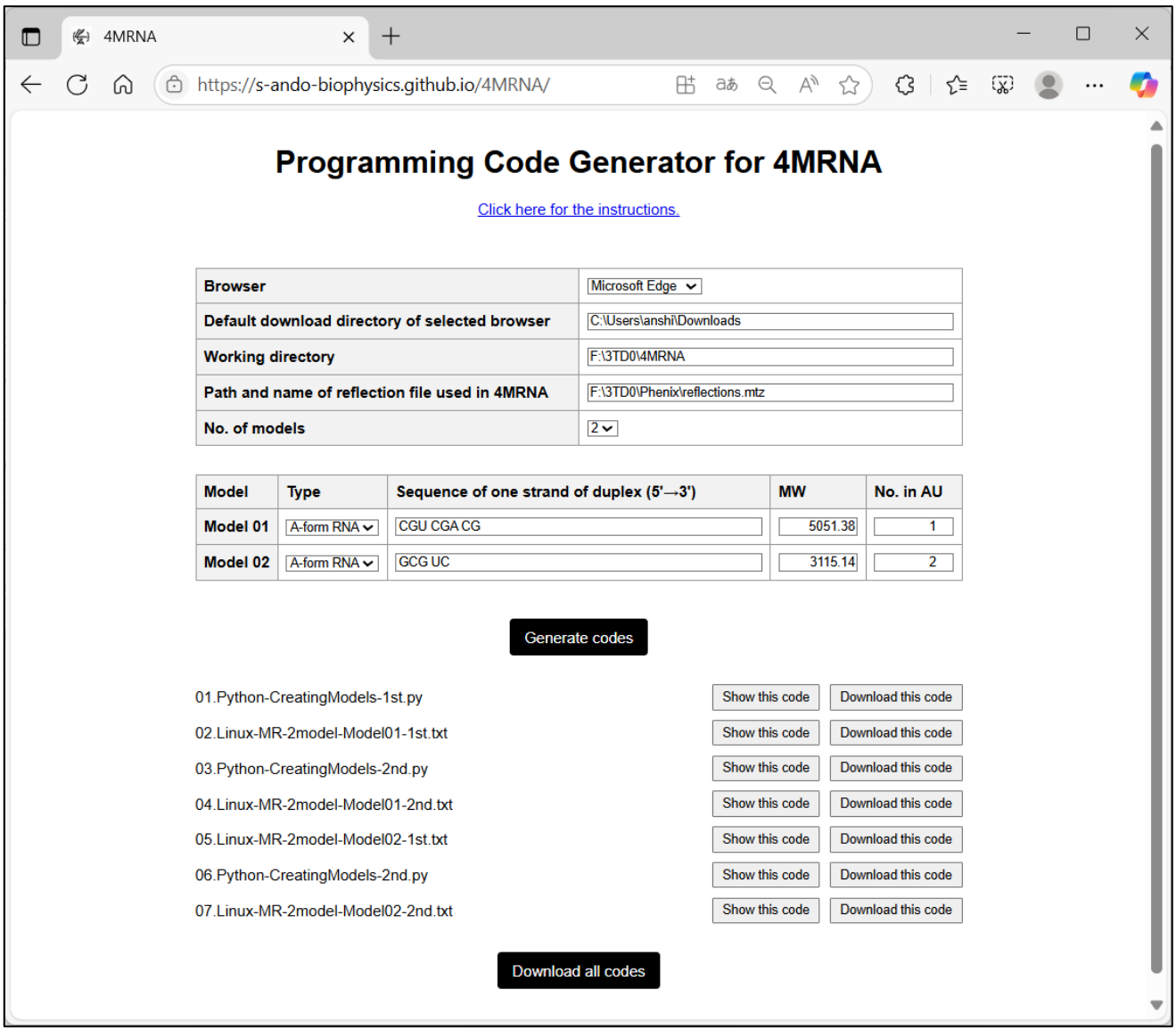
① Access the website "Programming Code Generator for 4MRNA".

② Enter the required information and complete the table. For details, refer to Detailed Instructions 【Entering Required Information】.

③ Once all required fields are filled in, click the "Generate codes" button. Programming codes corresponding to the entered information will be generated and displayed on the screen.

④ Click the "Download all codes" button to download all the codes.

⑤ The downloaded programming codes are numbered, so execute them in order in your environment. For detailed procedures, refer to Detailed Instructions 【Executing the Codes】.



Detailed Instructions

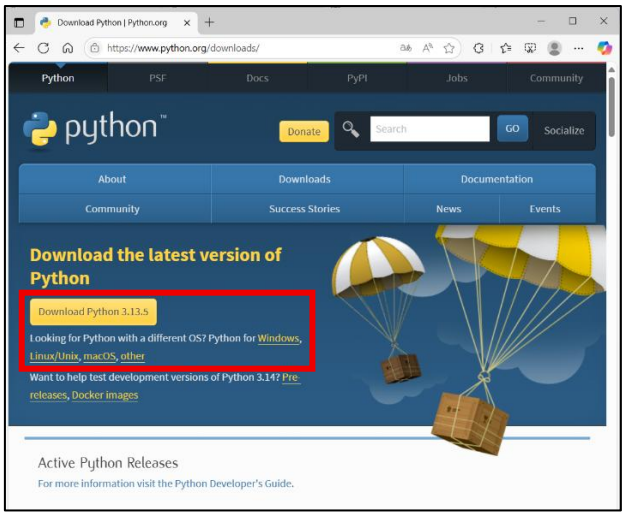
【Preparation and Setup for Using 4MRNA】

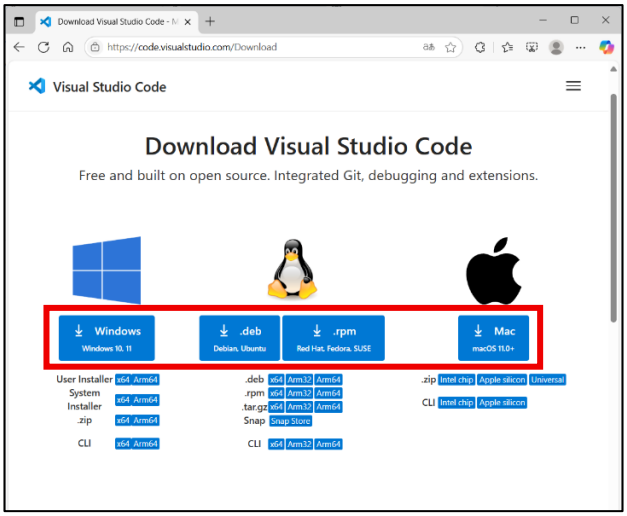
(1) Enable Python

① Install Python. Access the official website, download the installer suitable for your computer's operating system, and run it.

② After installation, open a terminal and execute the following command: pip install selenium pandas.

③ It is recommended (though not mandatory) to download a text editor like Visual Studio Code, which makes it easier to run Python codes.



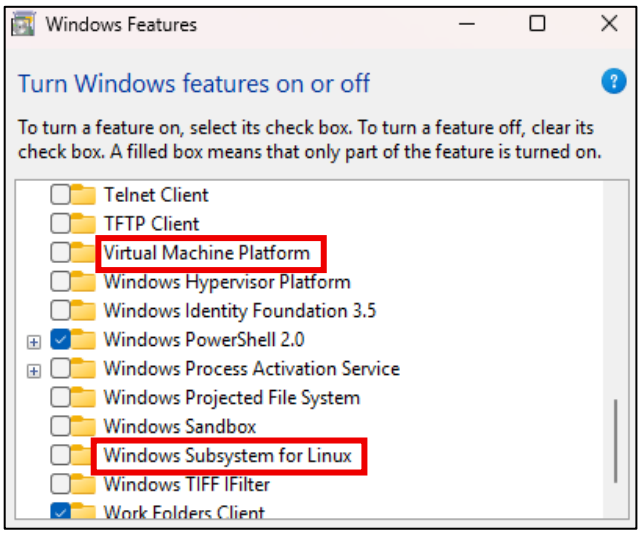


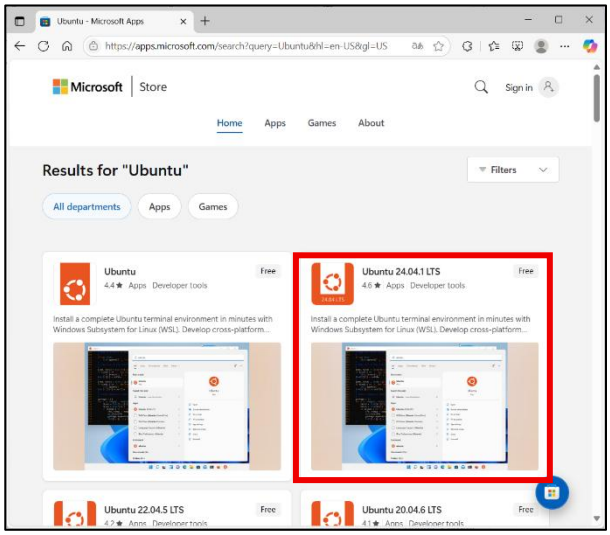
(2) Enable Shell Scripts (Windows Only)

① Enable "Windows Subsystem for Linux (WSL)" and "Virtual Machine Platform". Go to Control Panel > Programs > Programs and Features > Turn Windows features on or off, and check the boxes for "Windows Subsystem for Linux" and "Virtual Machine Platform".

② Download Ubuntu.

③ After the download is complete, launch Ubuntu and run the following commands: sudo apt update and sudo apt upgrade.

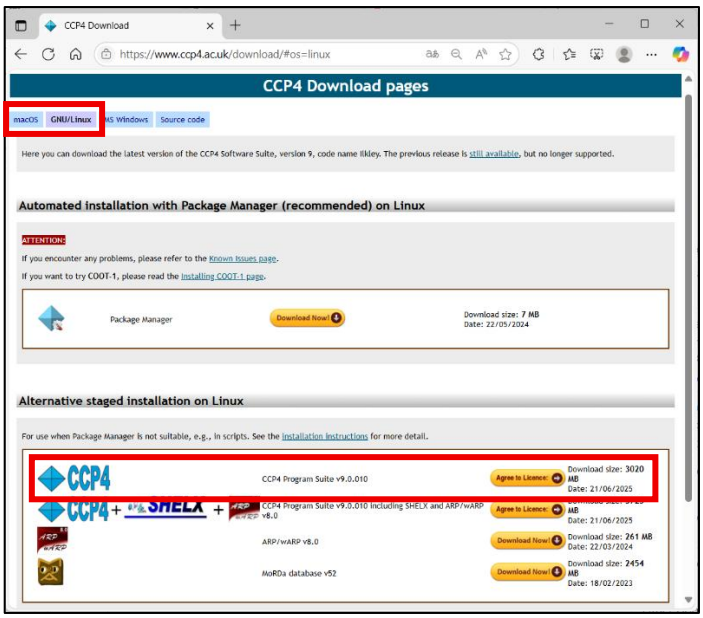




(3) Enable the Molecular Replacement Software *Phaser*

① Phaser is included in a software suite called CCP4. Visit the official website and download the macOS or Linux version (choose the Linux version even if you're using Windows).

② Once the download is complete, execute the following commands in order to complete the installation. The following commands are for Ubuntu on Windows, but similar commands should be used for macOS and Linux. (These commands assume that the file “ccp4-9.0.010-linux64.tar.gz” has been downloaded to “C:¥Users¥name¥Downloads”. You will need to adjust the version number and folder path as appropriate.)



- sudo su
- cd /usr/local
- mv /mnt/c/Users/name/Downloads/ccp4-9.0.010-linux64.tar.gz .
- gunzip ccp4-9.0.010-linux64.tar.gz
- tar -xvf ccp4-9.0.010-linux64.tar
- cd ccp4-9
- ./BINARY.setup
- exit
- cd /home/name
- vi .bashrc
- source /usr/local/ccp4-9/bin/ccp4.setup-sh
  # Please add to the last line.

(4) Others

① If you are using Safari (macOS only), an additional setting is required. Go to Safari Preferences > Develop, and check “Allow Remote Automation”.

② For checking the results of molecular replacement, it is convenient (though not essential) to have spreadsheet software like Microsoft Excel installed.

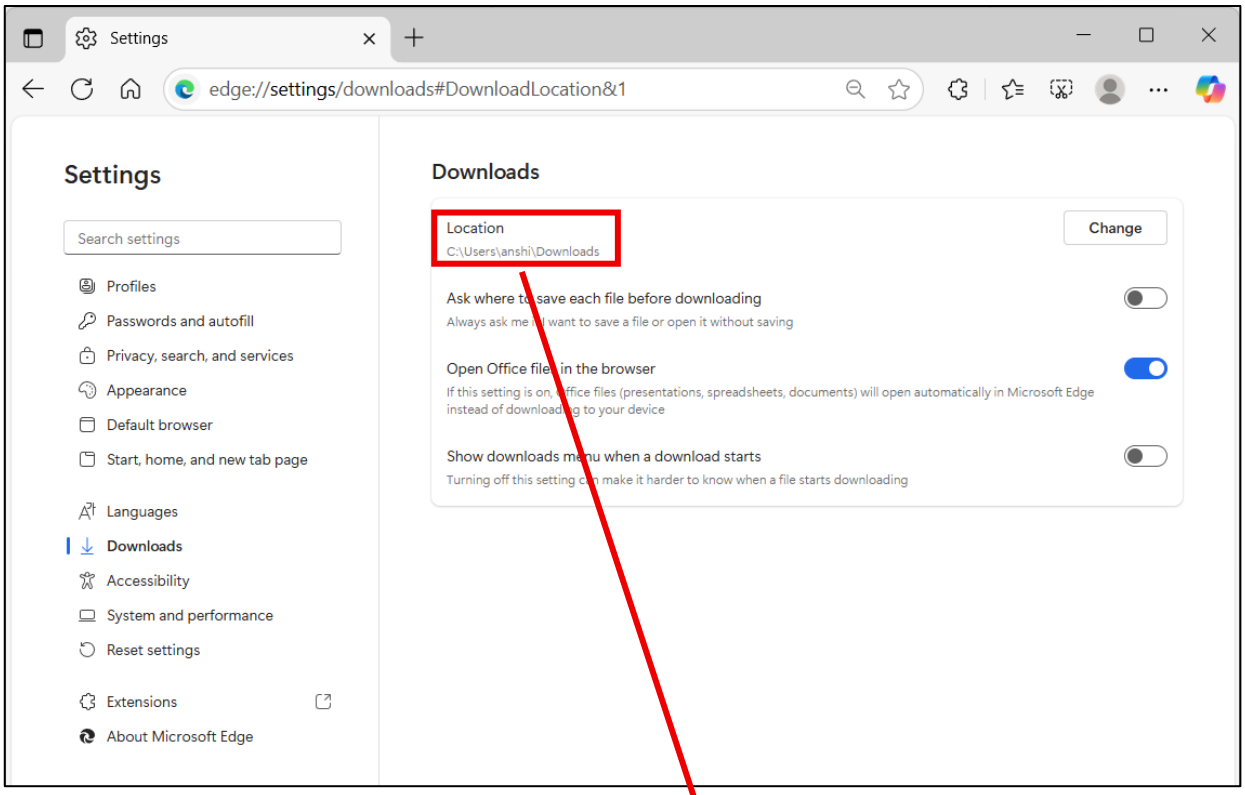
【Entering Required Information (Step②)】

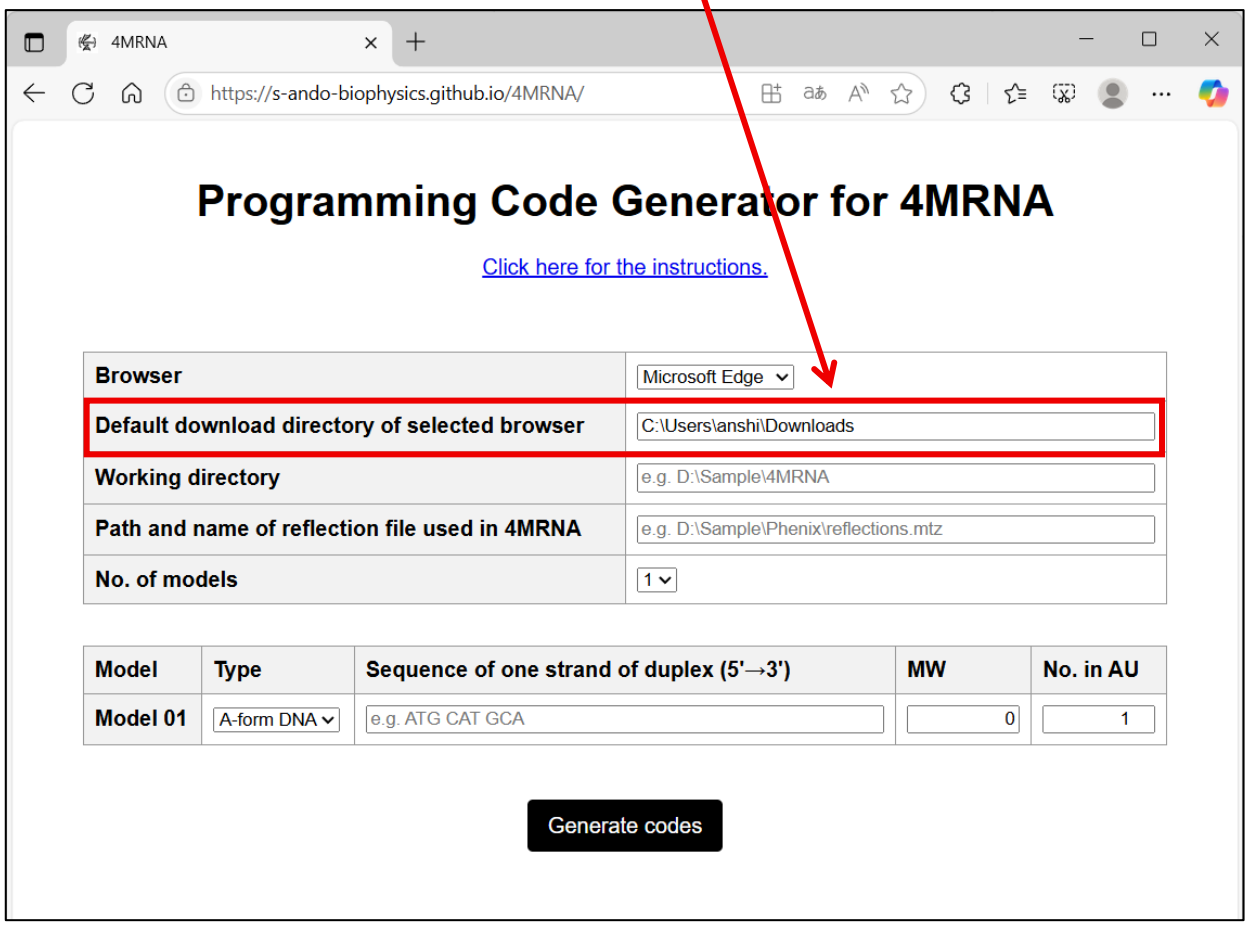
(1)	**Browser**	Select the browser you are using from the following options: Google Chrome, Microsoft Edge, Firefox, Safari.
(2)	**Default download directory of selected browser**	Open the settings of the browser selected in (1), confirm the default download directory, and enter it in this field.
(3)	**Working directory**	Create a directory where 4MRNA will operate (the output files will be saved here), and enter its full path.
(4)	**Path and name of reflection file used in 4MRNA**	Specify the full path and file name of the reflection file used for molecular replacement.

（2）How to Check the "Default download directory of selected browser"

■ In most browsers, it can be found under Settings > Downloads.

（例）Windows・Microsoft Edge

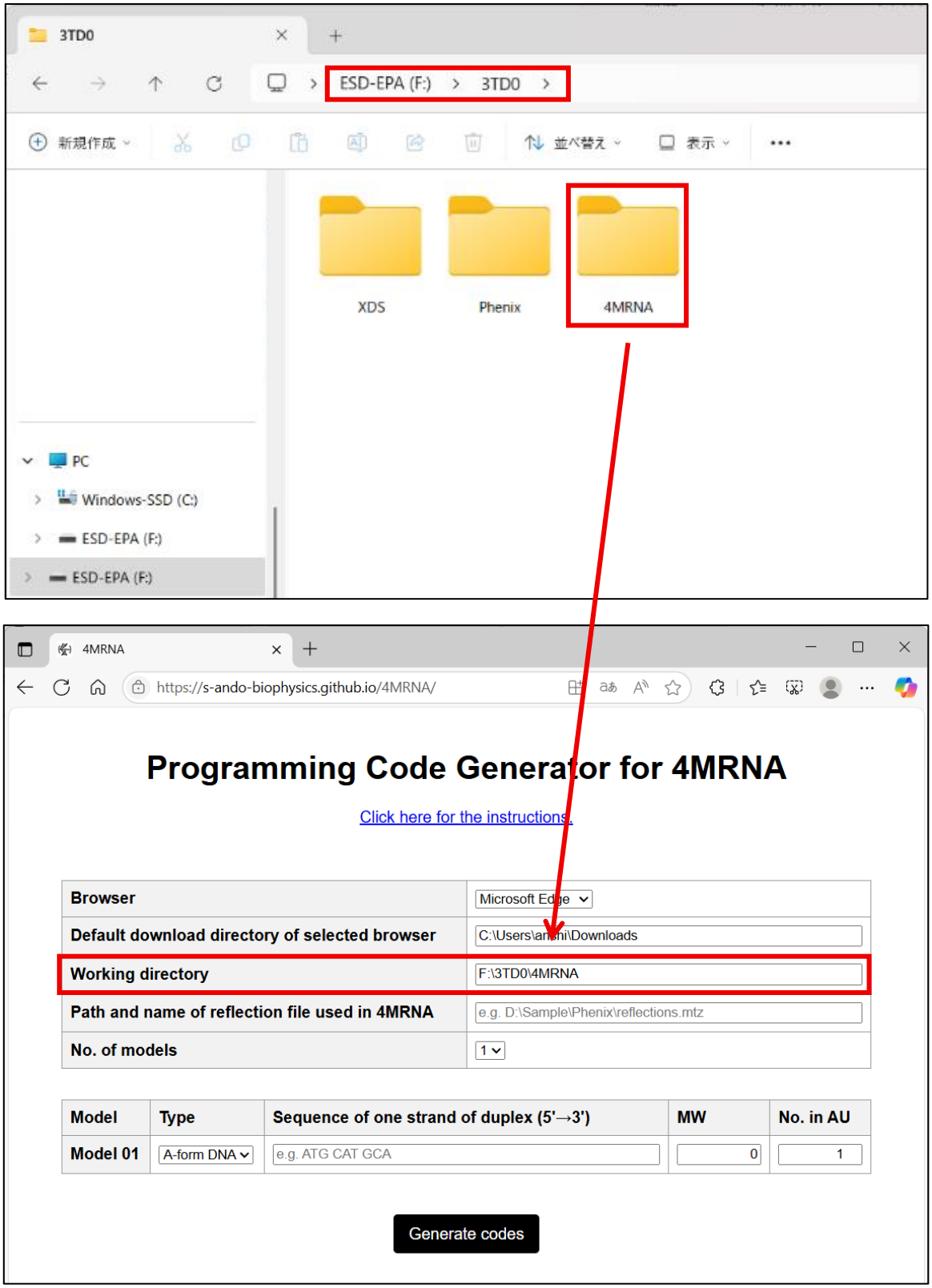




（3）**About the "Working directory"**

- Create a new folder that 4MRNA will use. Models, parameter files used in model generation, and molecular replacement results will be output to this folder.

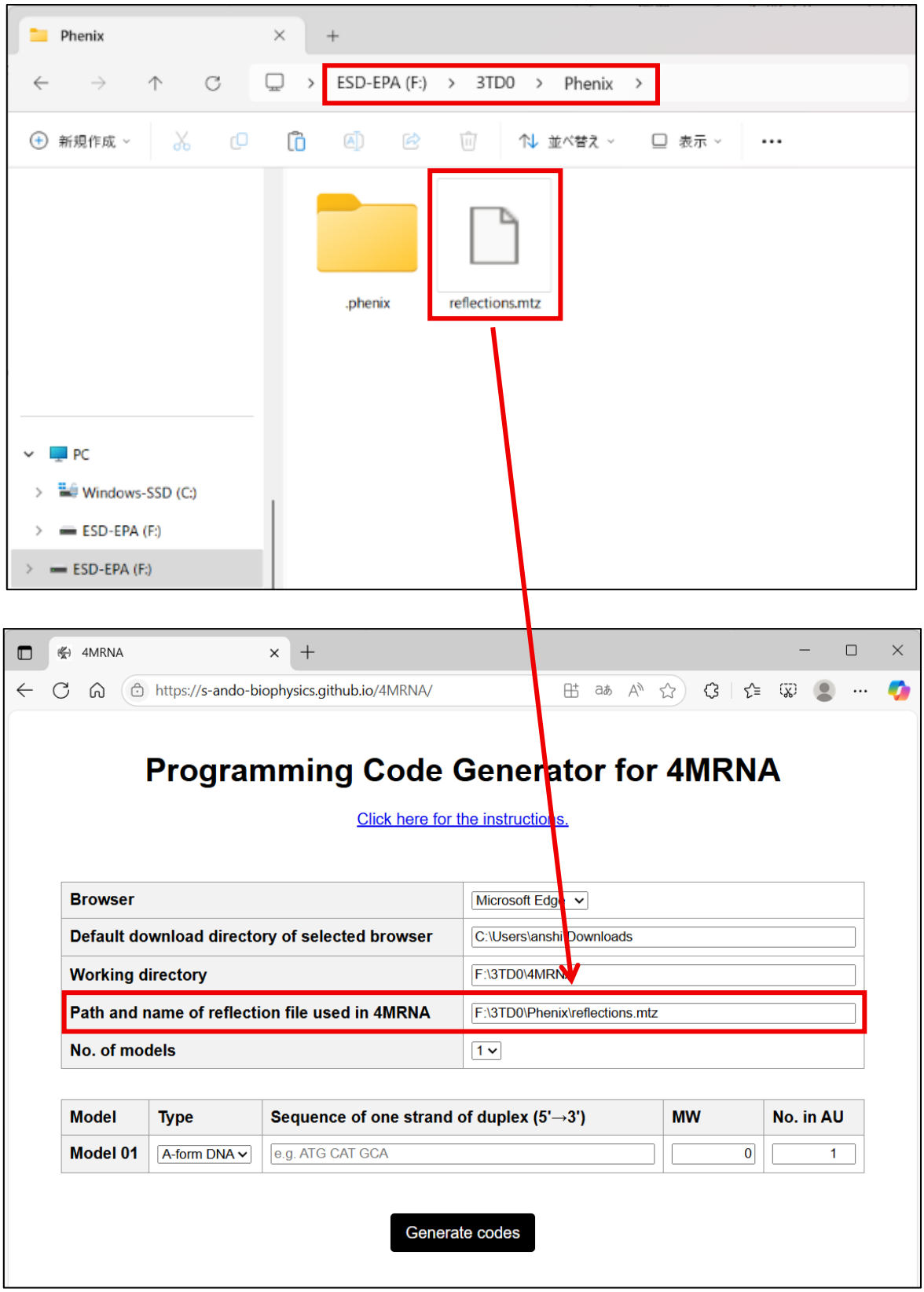
（例）Suppose your sample is named 3TD0, and the X-ray diffraction data is stored in a folder 3TD0 on a USB drive (F:). In that case, create a new folder named “4MRNA” inside “F:¥3TD0”, and input the path "F:¥3TD0¥4MRNA” on the website.



（4）**How to Check the "Path and name of reflection file used in 4MRNA”**

- You must specify the location and name of the reflection file required for molecular replacement. When using the software *Phenix*, the reflection file is typically named reflections.mtz.

（例）If “reflections.mtz” is located in “F:¥3TD0¥Phenix”, then input “F:¥3TD0¥Phenix¥reflections.mtz”.



(5)	**No. of models**	Select the number of types of search models to be used for molecular replacement.
(6)	**Model Information**	According to the number selected in (5), input fields will appear. For each model, enter the type of nucleic acid duplex, the sequence, and the number of models in the asymmetric unit.

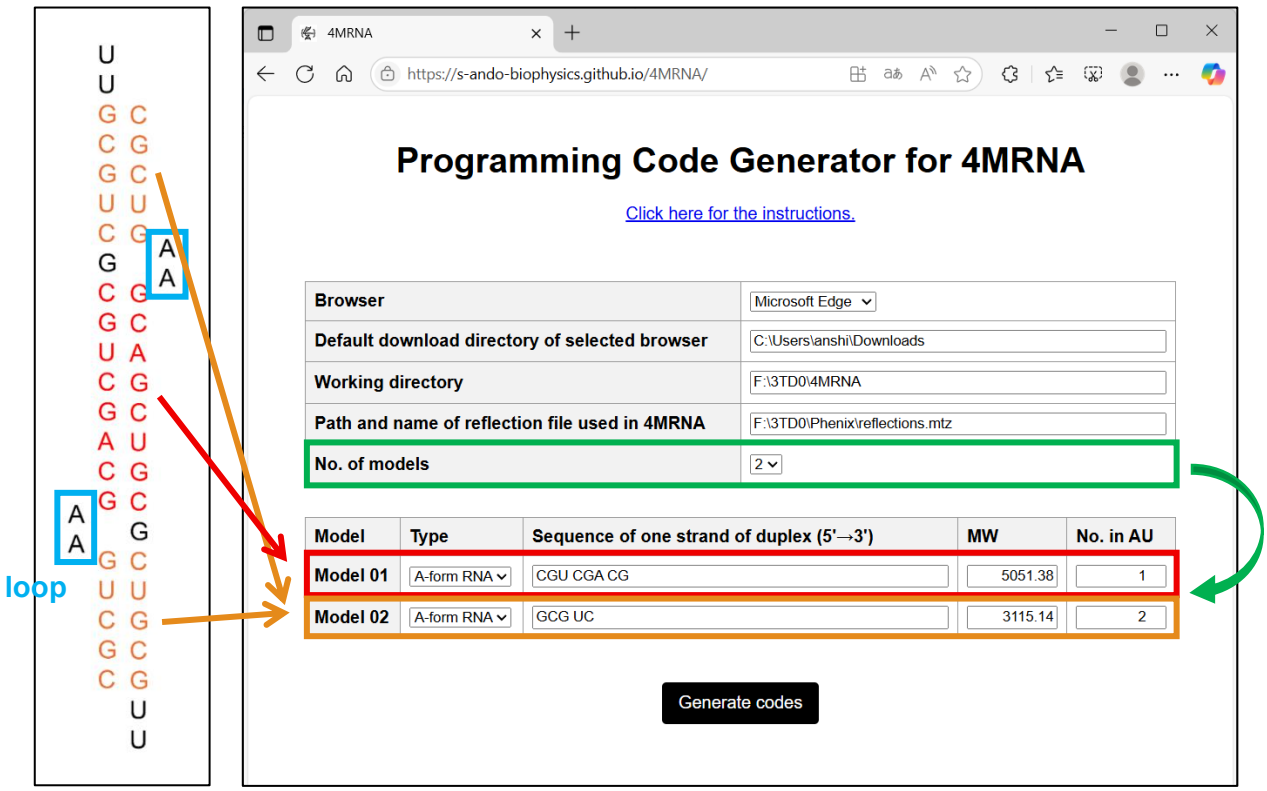
(5) (6) About Search Models Using for Molecular Replacement

① If your sample includes loops in the duplex, divide it into multiple stem regions to avoid the loops and create separate search models for each stem. In that case, the number of models you create (i.e., the number of stem regions) should match the value selected in (5).

② According to the number selected in (5), input fields will appear. For each model (stem), enter the following information.

- **Type**: Select the type of the duplex (stem) from A-form DNA, B-form DNA, or A-form RNA.
- **Sequence of one strand of duplex**: Input the base sequence (5'→3') of one strand of the duplex. The complementary strand will be automatically generated. (Even if mismatches exist, a fully complementary model will be created, which can be corrected later during refinement..)
- **MW**: The molecular weight of the duplex (including complementary strand) will be automatically calculated based on the input sequence.
- **No. in AU**: Enter how many copies of the model exist in the asymmetric unit. This value must be pre-estimated using indicators such as the Matthews coefficient.

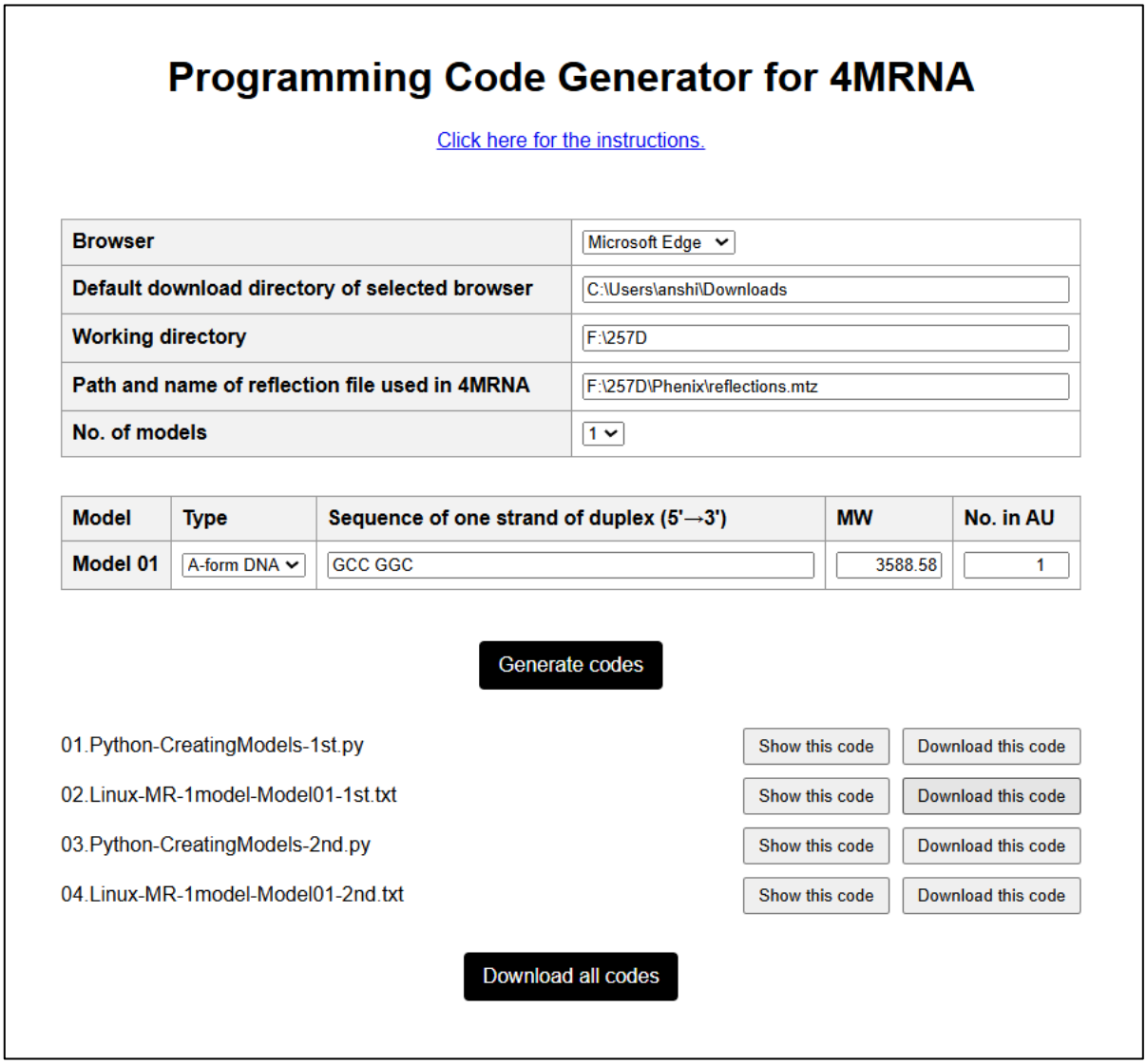
(例) Suppose the structure is expected to contain loops, as shown below. In this case, divide the model into one long stem (red) and two short stems (orange), and input their information accordingly. Assume the Matthews coefficient suggests that one copy of the entire molecule (long stem + 2 short stems + loops) is present in the asymmetric unit.



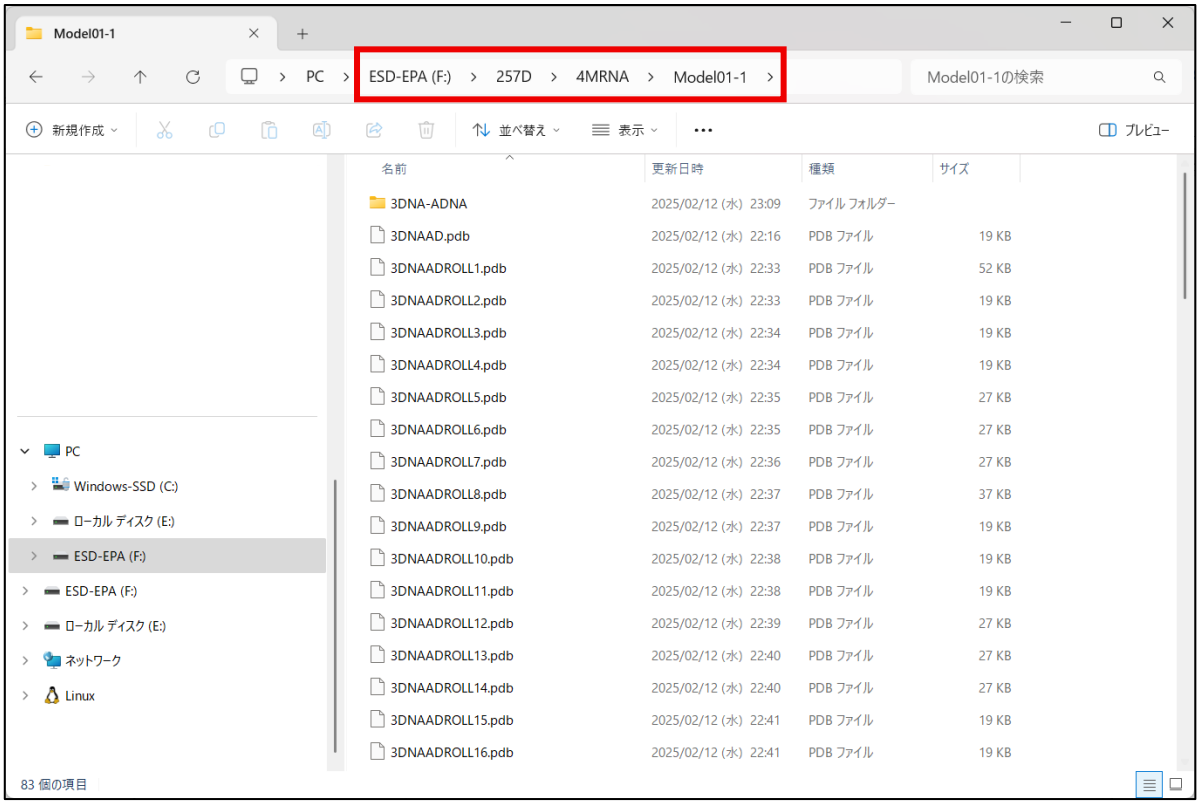
【Executing the Codes (Step⑤)】

Python code	Among the downloaded files, the ones that start with "Python" and have the .py extension are the Python codes. Open these files using Visual Studio Code or a similar text editor, and run them.
Shell script	Among the downloaded files, the ones that start with "Linux" and have the .txt extension are the shell scripts. Copy the entire content of these .txt files and paste it into Ubuntu (Windows) or a terminal (macOS/Linux).

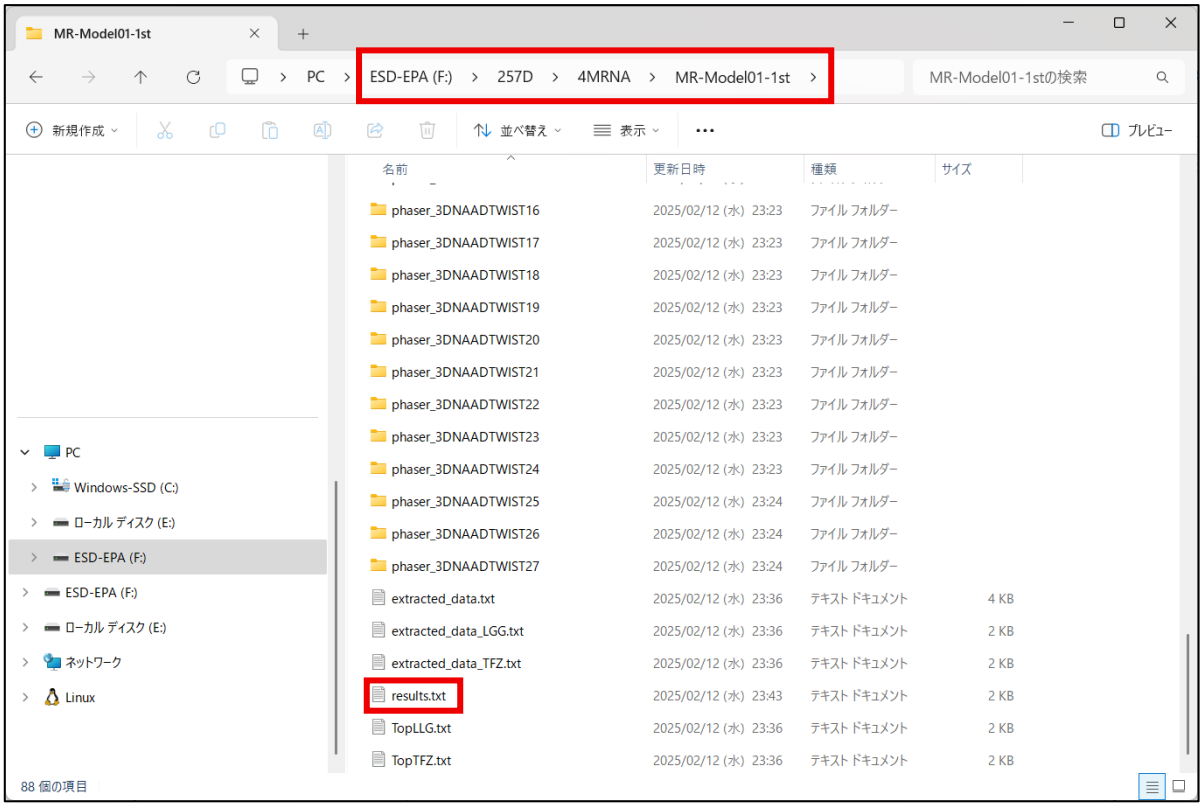
◆ If “No. of models” = 1



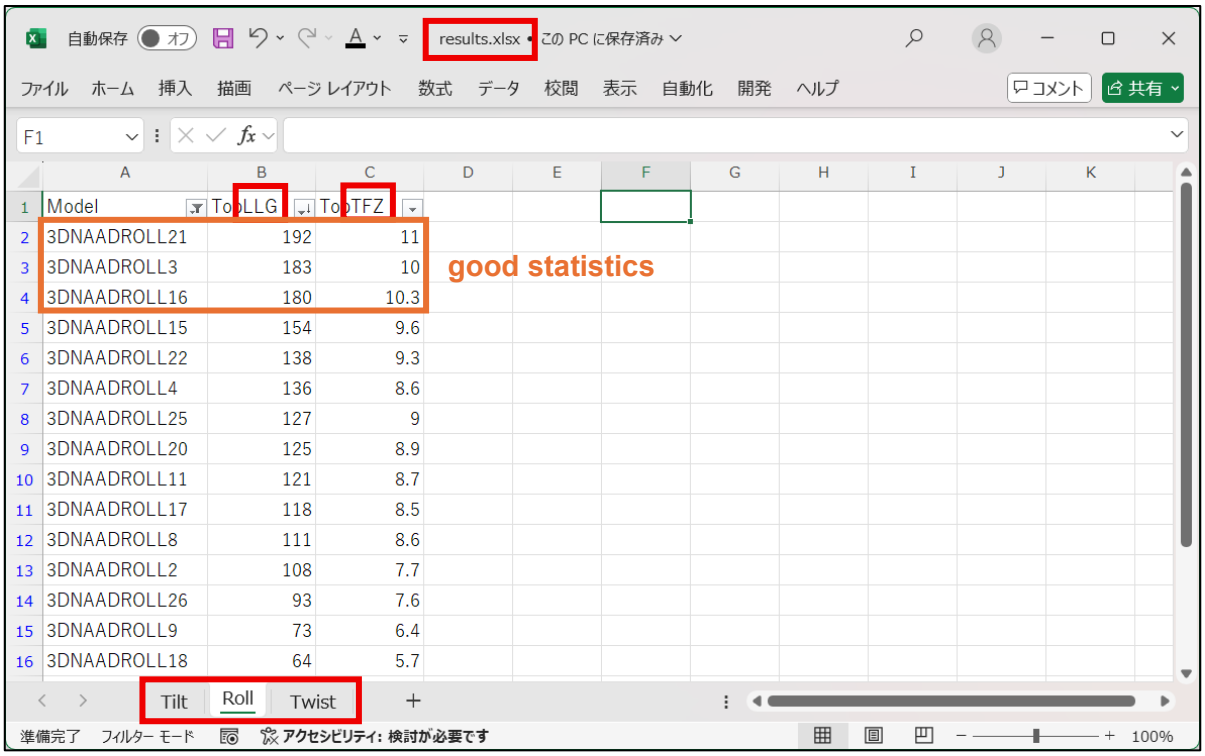
① Run “**01.Python-CreatingModels-1st.py**”. After execution, search models for molecular replacement will be created and downloaded into the folder 4MRNA > Model01-1.



② Run “**02.Linux-MR-1model-Model01-1st.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model01-1st.

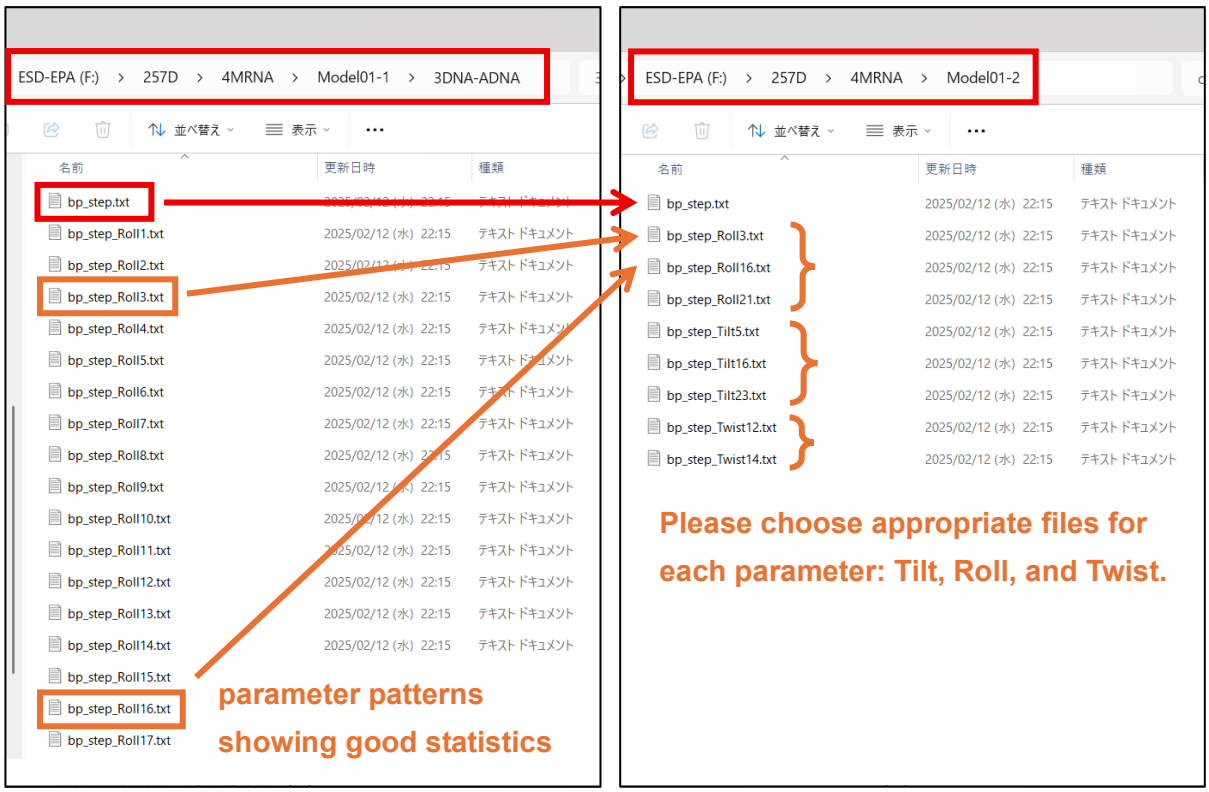


③ Open “results.txt” in Microsoft Excel and check parameter patterns for each of the three parameters (Tilt, Roll, Twist) that show good statistics (LLG and TFZ). Empirically, LLG ≥ 100 and TFZ ≥ 7 are typical thresholds.

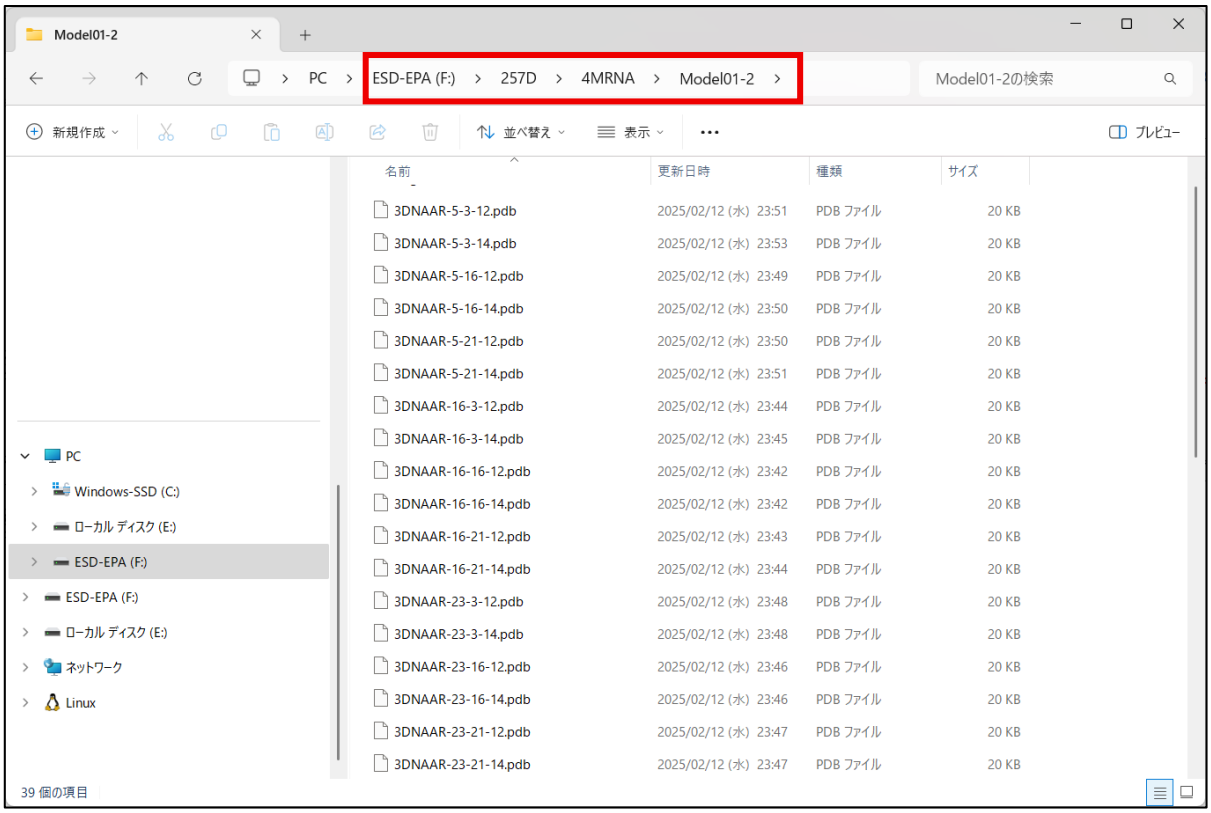


Model	TopLLG	TopTFZ
3DNAADROLL21	192	11
3DNAADROLL3	183	10
3DNAADROLL16	180	10.3
3DNAADROLL15	154	9.6
3DNAADROLL22	138	9.3
3DNAADROLL4	136	8.6
3DNAADROLL25	127	9
3DNAADROLL20	125	8.9
3DNAADROLL11	121	8.7
3DNAADROLL17	118	8.5
3DNAADROLL8	111	8.6
3DNAADROLL2	108	7.7
3DNAADROLL26	93	7.6
3DNAADROLL9	73	6.4
3DNAADROLL18	64	5.7

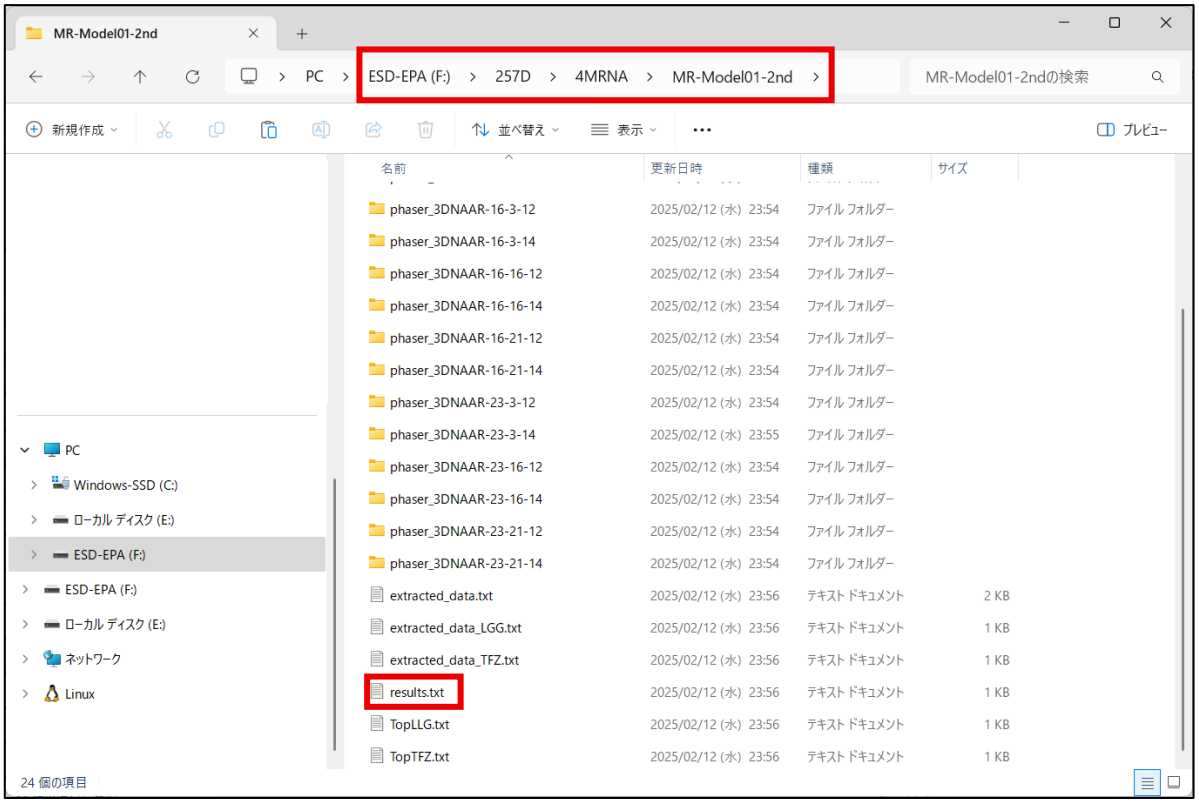
④ Go to the folder 4MRNA > Model01-1 > 3DNA-○○NA, find the files corresponding to the selected parameter patterns, and copy them into 4MRNA > Model01-2. Also copy the file named “bp_step.txt” into the same folder 4MRNA > Model01-2.



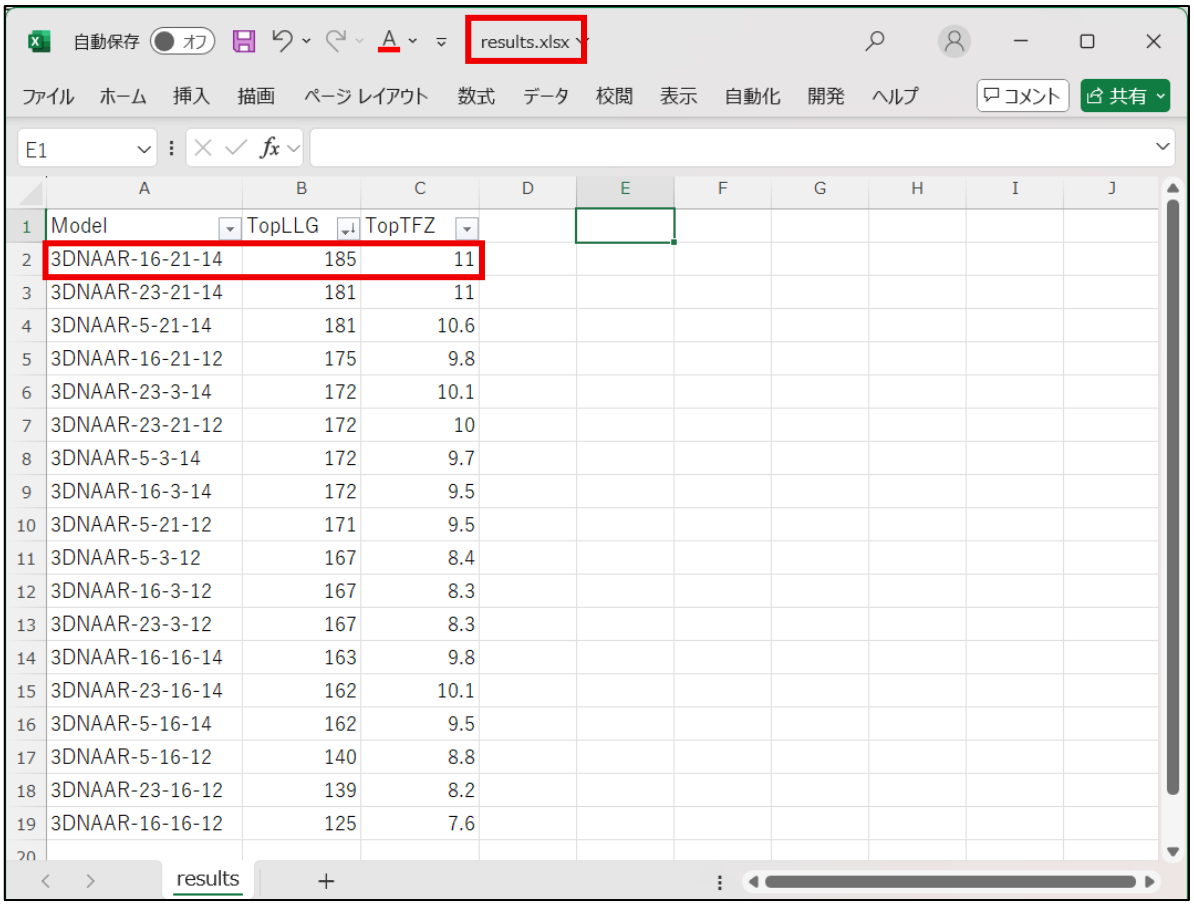
⑤ Run “**03.Python-CreatingModels-2nd.py**”. After execution, new models optimized by simultaneously adjusting Tilt, Roll, and Twist will be created in 4MRNA > Model01-2.



⑥ Run “**04.Linux-MR-1model-Model01-2nd.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model01-2nd.

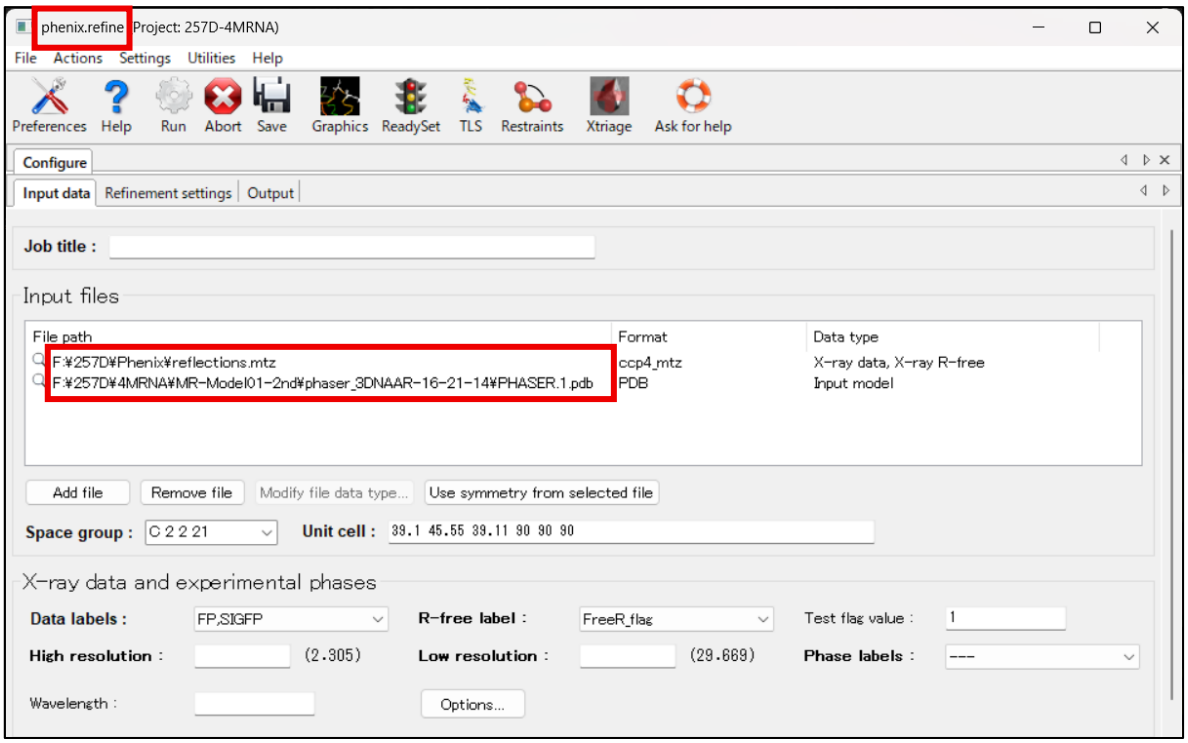


⑦ Open "results.txt" in Microsoft Excel to check the statistics and consider the one with the best value as the solution. In addition, visually validate the fit between the structure and the electron density map using a program like *Coot*.



Model	TopLLG	TopTFZ
3DNAAR-16-21-14	185	11
3DNAAR-23-21-14	181	11
3DNAAR-5-21-14	181	10.6
3DNAAR-16-21-12	175	9.8
3DNAAR-23-3-14	172	10.1
3DNAAR-23-21-12	172	10
3DNAAR-5-3-14	172	9.7
3DNAAR-16-3-14	172	9.5
3DNAAR-5-21-12	171	9.5
3DNAAR-5-3-12	167	8.4
3DNAAR-16-3-12	167	8.3
3DNAAR-23-3-12	167	8.3
3DNAAR-16-16-14	163	9.8
3DNAAR-23-16-14	162	10.1
3DNAAR-5-16-14	162	9.5
3DNAAR-5-16-12	140	8.8
3DNAAR-23-16-12	139	8.2
3DNAAR-16-16-12	125	7.6

⑧ After the 4MRNA is completed, further refinement is carried out using *Phenix Refine*, and the structure is finally determined.



◆ **If “No. of models” = 2**

Programming Code Generator for 4MRNA

Click here for the instructions.

Browser	Microsoft Edge
Default download directory of selected browser	C:\Users\anshi\Downloads
Working directory	F:\1T0E\4MRNA
Path and name of reflection file used in 4MRNA	F:\1T0E\Phenix\reflections.mtz
No. of models	2

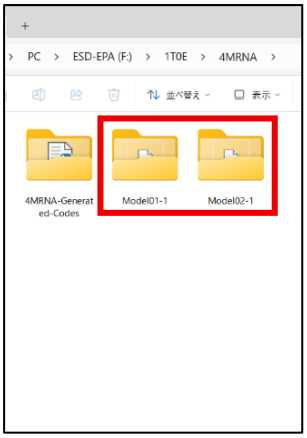
Model	Type	Sequence of one strand of duplex (5'→3')	MW	No. in AU
Model 01	A-form RNA	GUC GCU CG	5051.38	1
Model 02	A-form RNA	GGU GGU G	4400.96	1

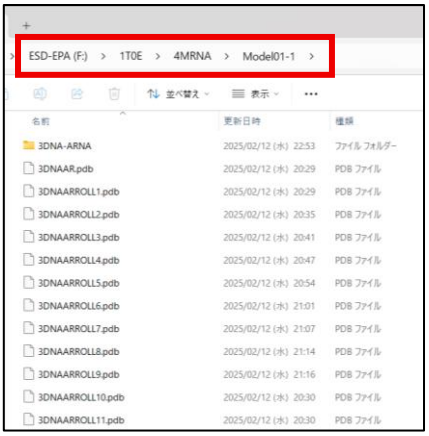
Generate codes

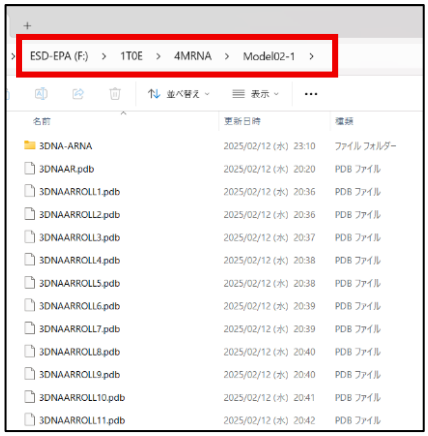
01.Python-CreatingModels-1st.py	Show this code	Download this code
02.Linux-MR-2model-Model01-1st.txt	Show this code	Download this code
03.Python-CreatingModels-2nd.py	Show this code	Download this code
04.Linux-MR-2model-Model01-2nd.txt	Show this code	Download this code
05.Linux-MR-2model-Model02-1st.txt	Show this code	Download this code
06.Python-CreatingModels-2nd.py	Show this code	Download this code
07.Linux-MR-2model-Model02-2nd.txt	Show this code	Download this code

Download all codes

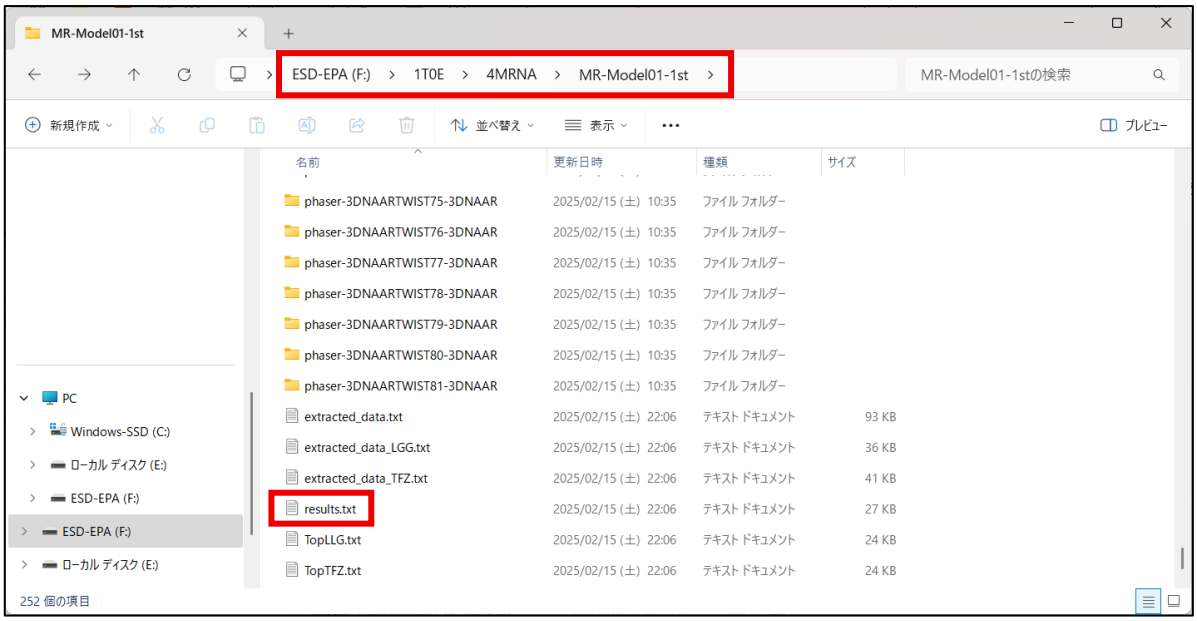
① Run “**01.Python-CreatingModels-1st.py**”. After execution, the first search models will be downloaded into the folder 4MRNA > Model01-1, and the second search models into 4MRNA > Model02-1.



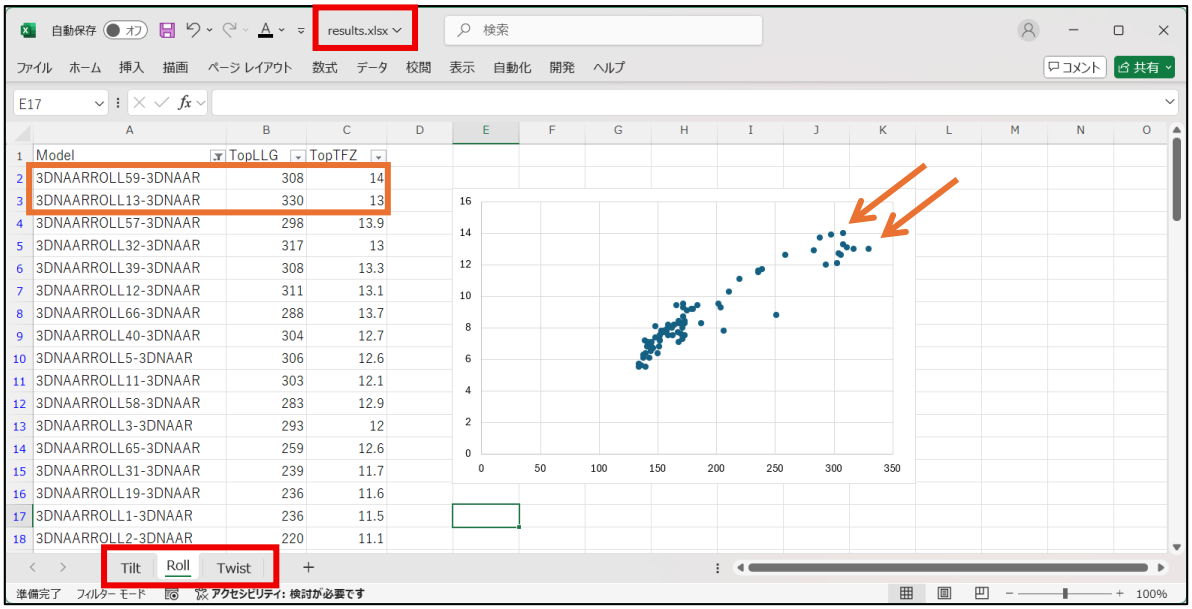




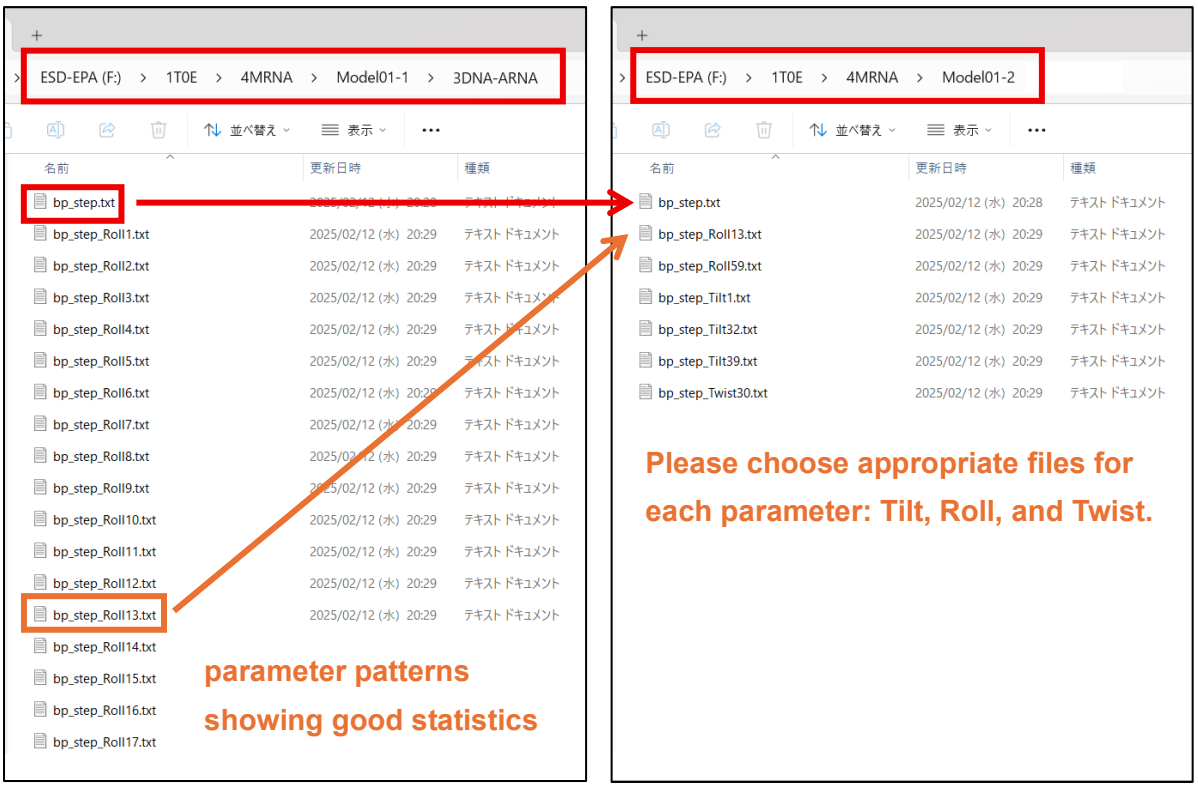
② Run “**02.Linux-MR-2model-Model01-1st.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model01-1st.



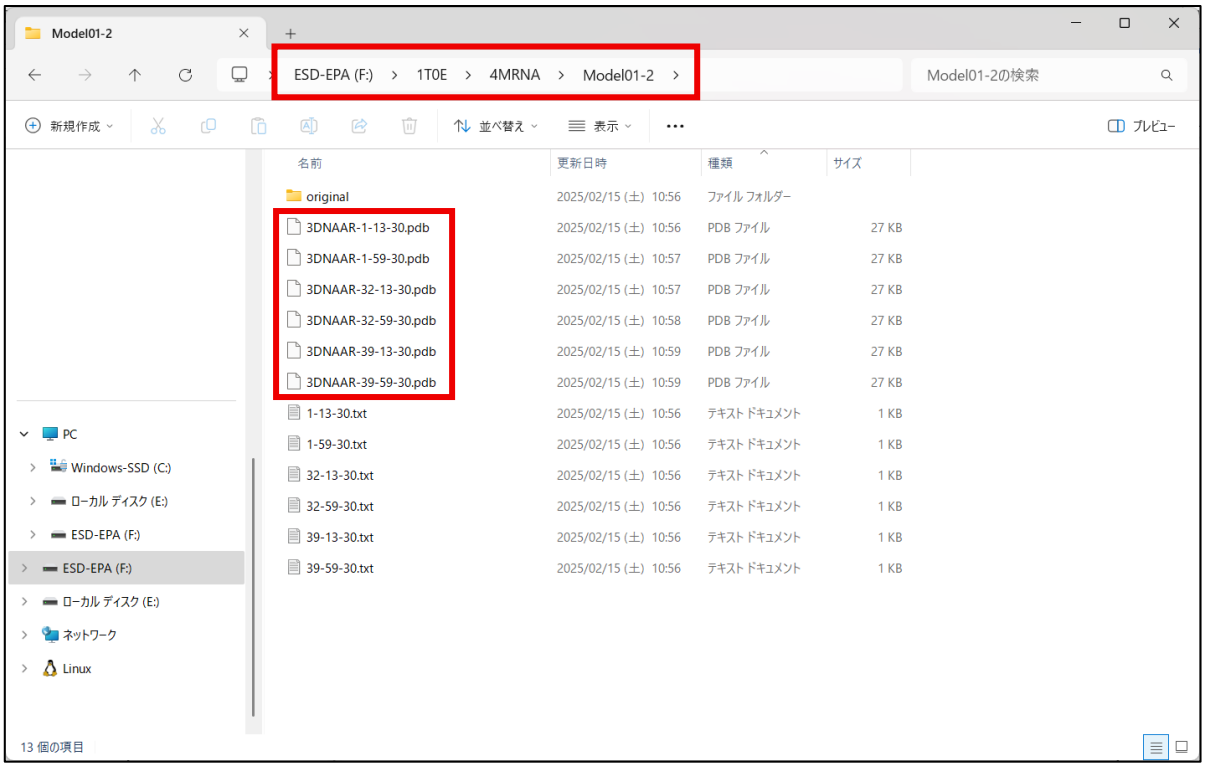
③ Open “results.txt” in Microsoft Excel and check parameter patterns for each of the three parameters (Tilt, Roll, Twist) that show good statistics (LLG and TFZ). Empirically, LLG ≥ 100 and TFZ ≥ 7 are typical thresholds.



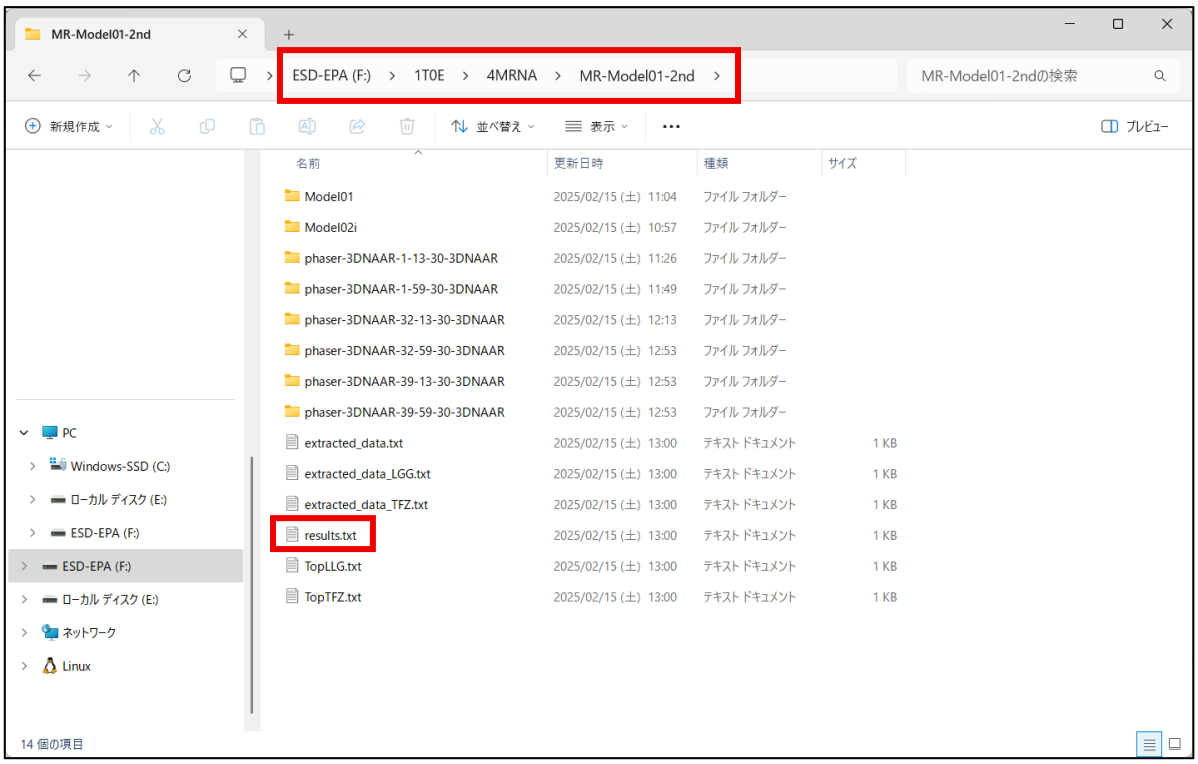
④ Go to the folder 4MRNA > Model01-1 > 3DNA-○○NA, find the files corresponding to the selected parameter patterns, and copy them into 4MRNA > Model01-2. Also copy the file named “bp_step.txt” into the same folder 4MRNA > Model01-2.



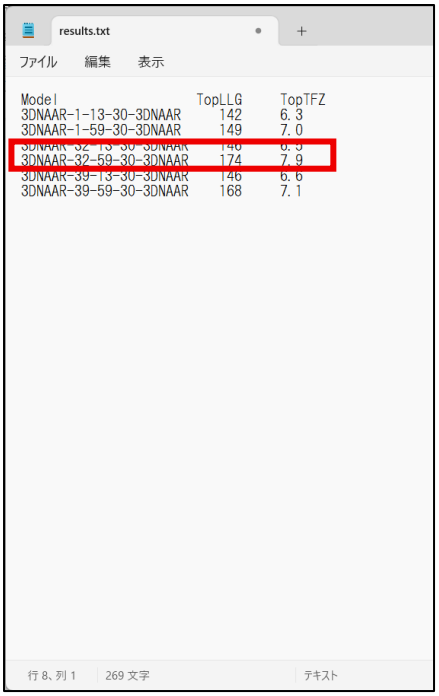
⑤ Run “**03.Python-CreatingModels-2nd.py**”. After execution, new models optimized by simultaneously adjusting Tilt, Roll, and Twist will be created in 4MRNA > Model01-2.

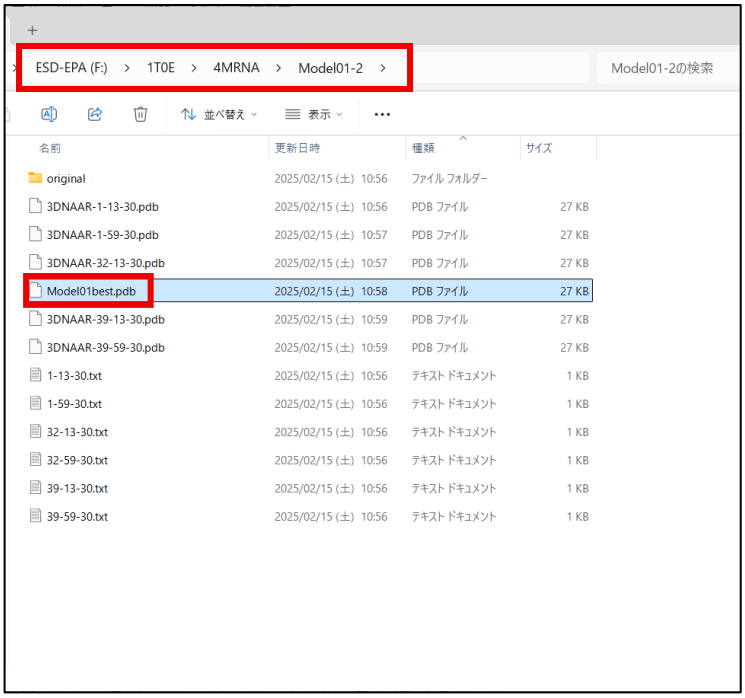


⑥ Run “**04.Linux-MR-2model-Model01-2nd.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model01-2nd.

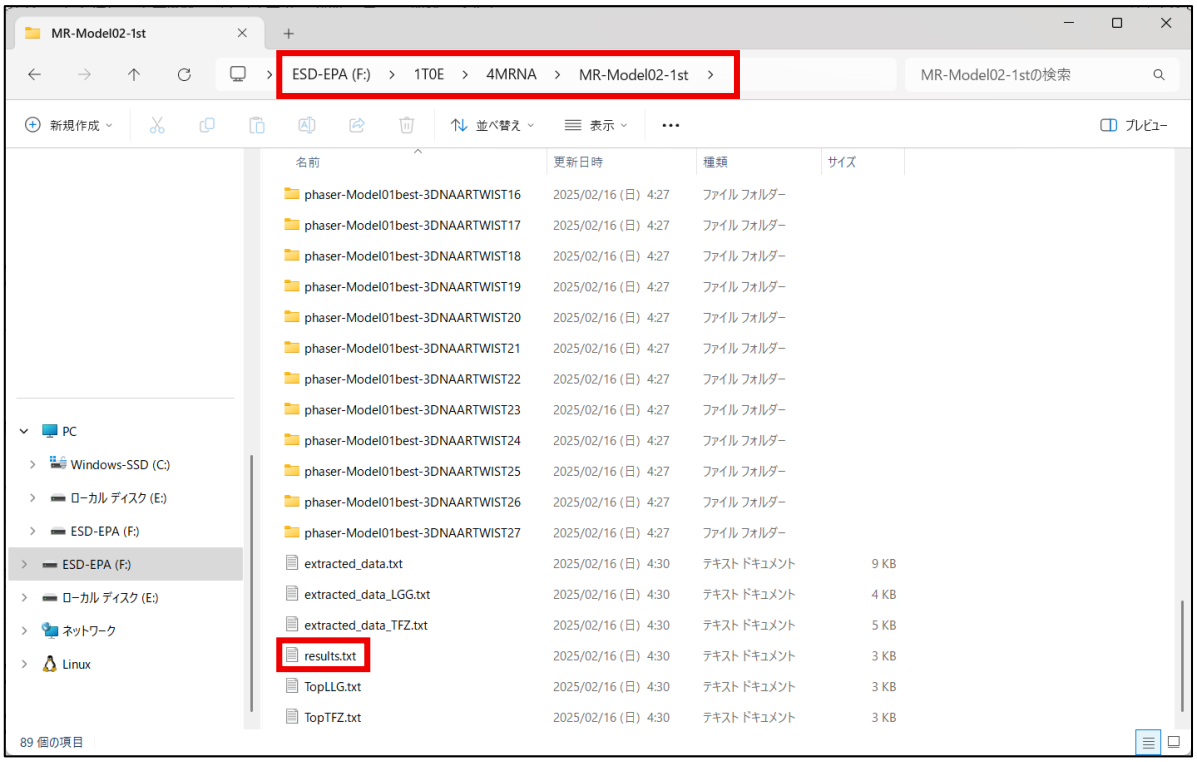


⑦ Open "results.txt" in Microsoft Excel to check the statistics and consider the one with the best value as the solution for the first model. Then rename the corresponding PDB file to “Model01best.pdb”

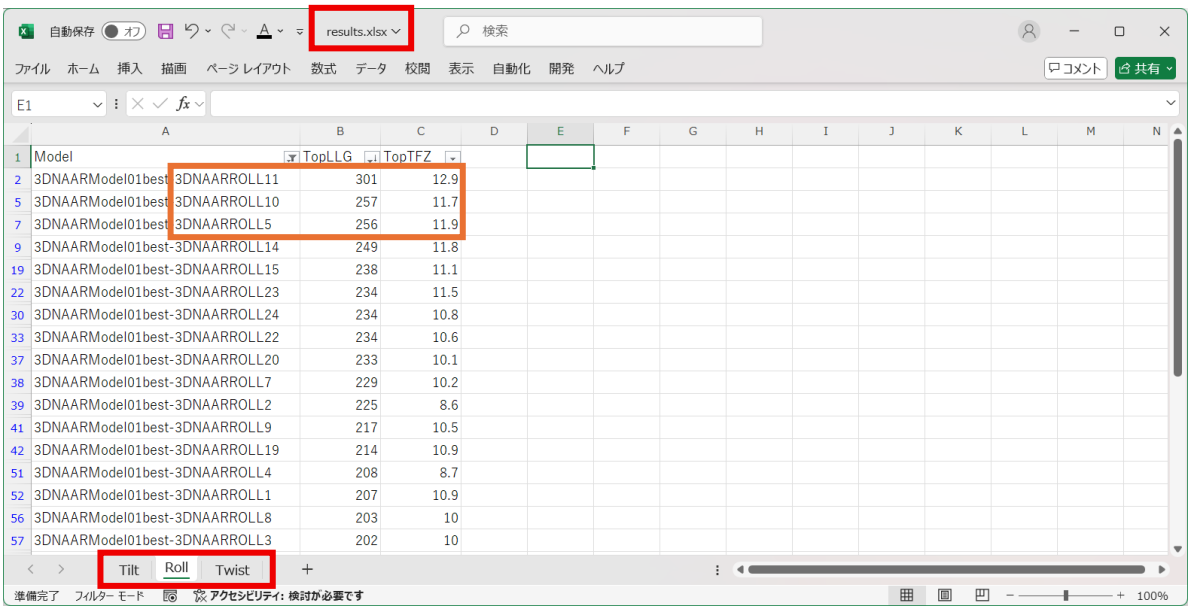




⑧ Run “**05.Linux-MR-2model-Model02-1st.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model02-1st.

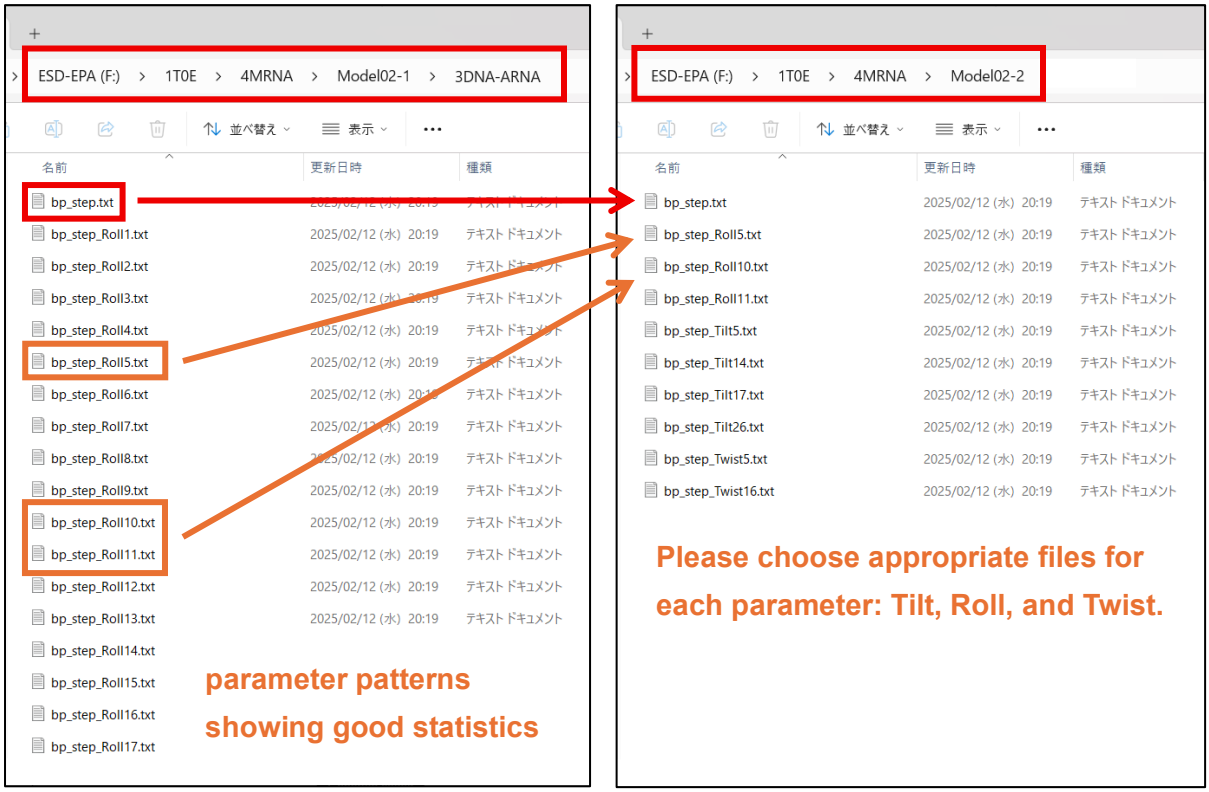


⑨ Open “results.txt” in Microsoft Excel and check parameter patterns for each of the three parameters (Tilt, Roll, Twist) that show good statistics (LLG and TFZ). Empirically, LLG ≥ 100 and TFZ ≥ 7 are typical thresholds.

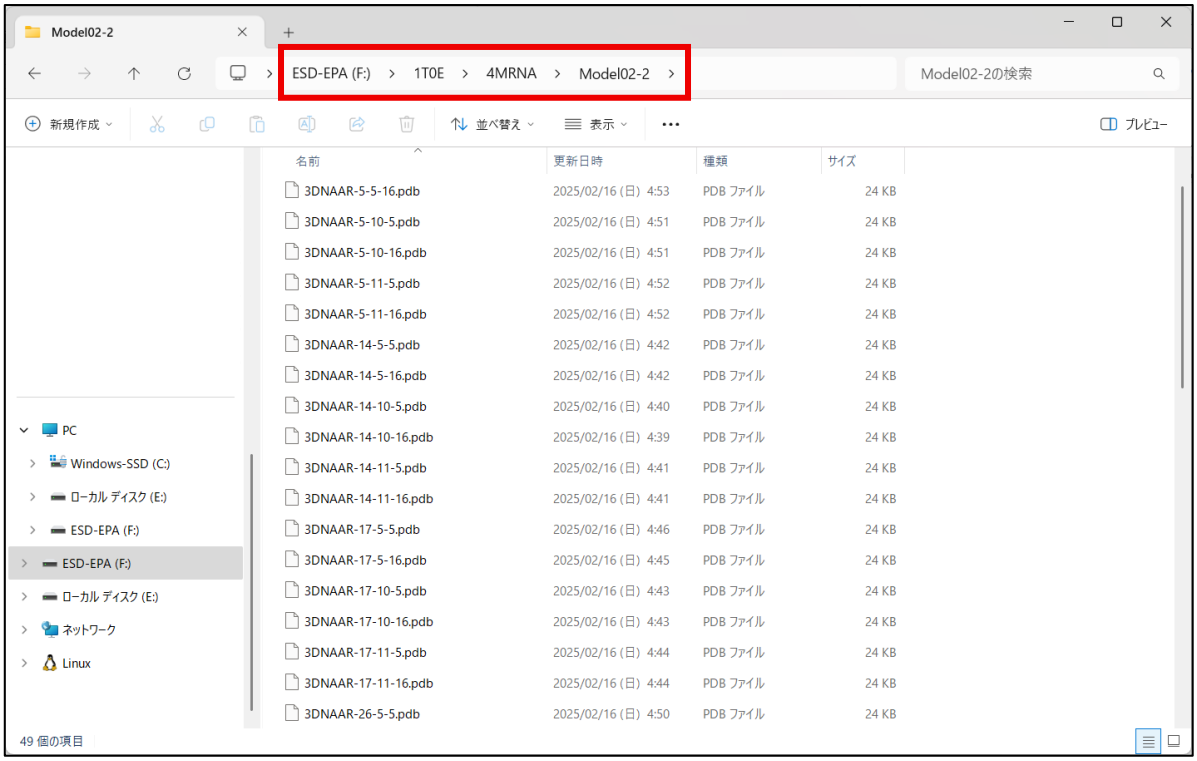


Model	TopLLG	TopTFZ
3DNAARModel01best-3DNAARROLL11	301	12.9
3DNAARModel01best-3DNAARROLL10	257	11.7
3DNAARModel01best-3DNAARROLL5	256	11.9
3DNAARModel01best-3DNAARROLL14	249	11.8
3DNAARModel01best-3DNAARROLL15	238	11.1
3DNAARModel01best-3DNAARROLL23	234	11.5
3DNAARModel01best-3DNAARROLL24	234	10.8
3DNAARModel01best-3DNAARROLL22	234	10.6
3DNAARModel01best-3DNAARROLL20	233	10.1
3DNAARModel01best-3DNAARROLL7	229	10.2
3DNAARModel01best-3DNAARROLL2	225	8.6
3DNAARModel01best-3DNAARROLL9	217	10.5
3DNAARModel01best-3DNAARROLL19	214	10.9
3DNAARModel01best-3DNAARROLL4	208	8.7
3DNAARModel01best-3DNAARROLL1	207	10.9
3DNAARModel01best-3DNAARROLL8	203	10
3DNAARModel01best-3DNAARROLL3	202	10

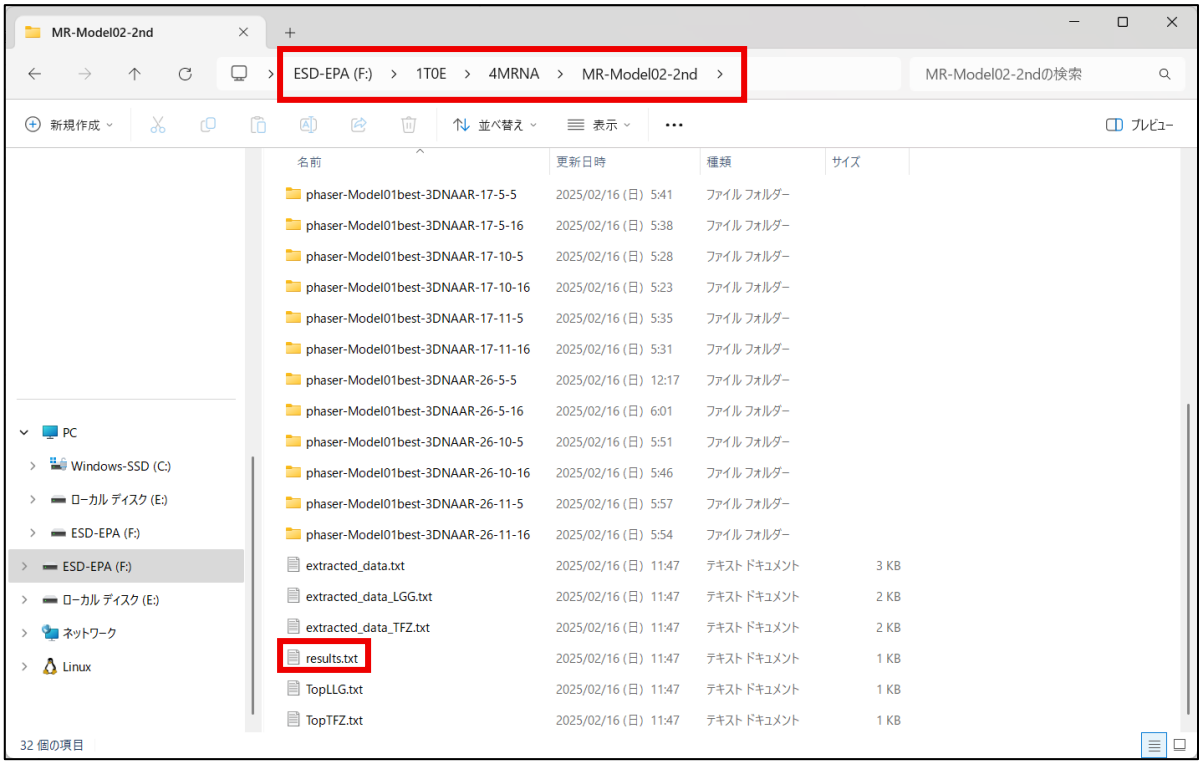
⑩ Go to the folder 4MRNA > Model02-1 > 3DNA-○○NA, find the files corresponding to the selected parameter patterns, and copy them into 4MRNA > Model02-2. Also copy the file named “bp_step.txt” into the same folder 4MRNA > Model02-2.



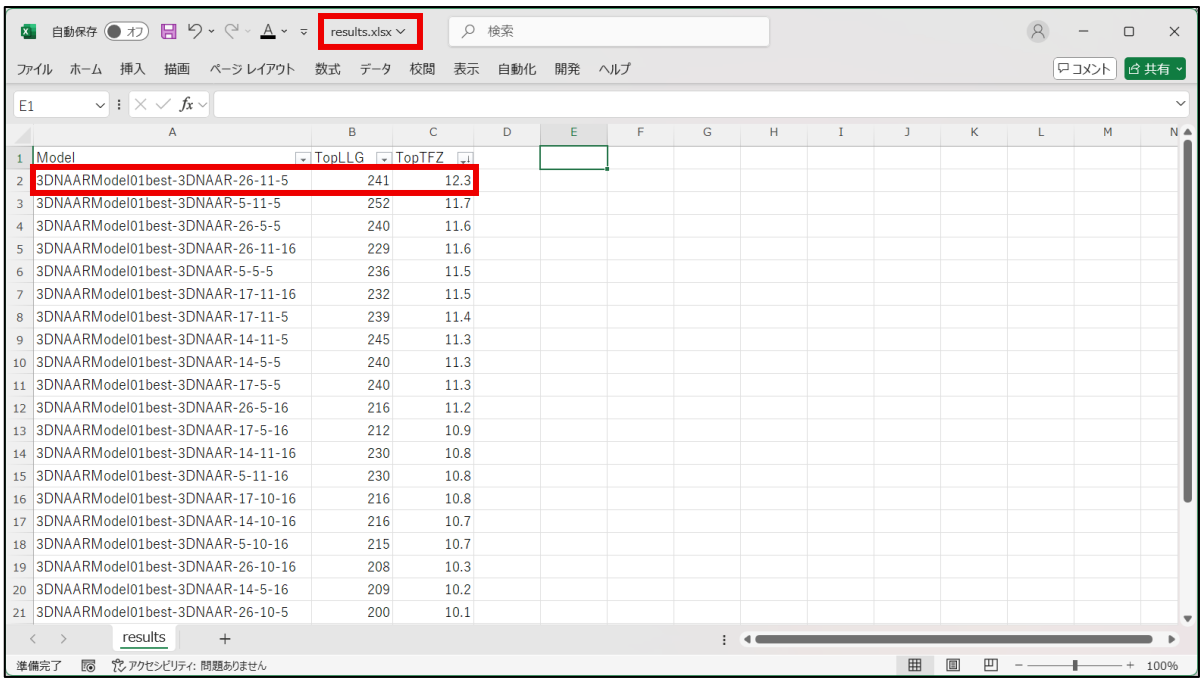
⑪ Run “**06.Python-CreatingModels-2nd.py**”. After execution, new models optimized by simultaneously adjusting Tilt, Roll, and Twist will be created in 4MRNA > Model02-2.



⑫ Run “**07.Linux-MR-2model-Model02-2nd.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model02-2nd.

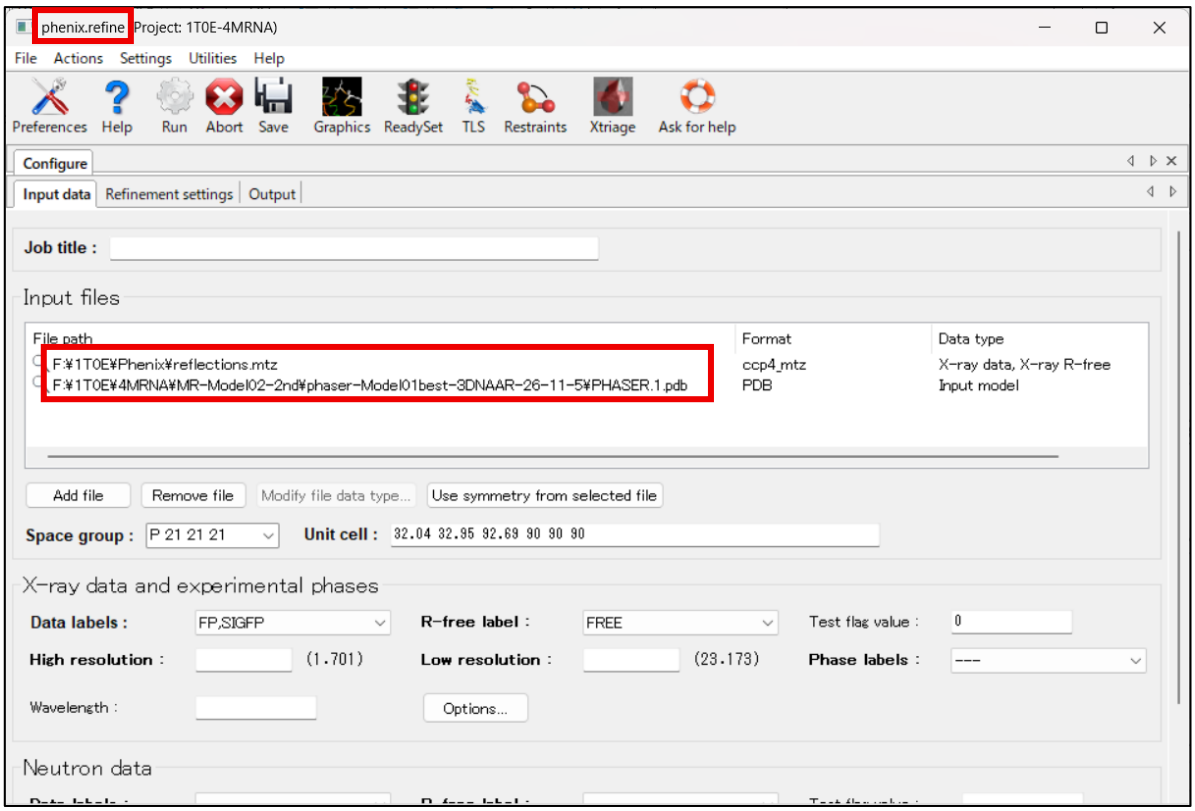


⑬ Open "results.txt" in Microsoft Excel to check the statistics and consider the one with the best value as the solution for the second model. In addition, visually validate the fit between the structure and the electron density map using a program like *Coot*.



Model	TopLLG	TopTFZ
3DNAARModel01best-3DNAAR-26-11-5	241	12.3
3DNAARModel01best-3DNAAR-5-11-5	252	11.7
3DNAARModel01best-3DNAAR-26-5-5	240	11.6
3DNAARModel01best-3DNAAR-26-11-16	229	11.6
3DNAARModel01best-3DNAAR-5-5-5	236	11.5
3DNAARModel01best-3DNAAR-17-11-16	232	11.5
3DNAARModel01best-3DNAAR-17-11-5	239	11.4
3DNAARModel01best-3DNAAR-14-11-5	245	11.3
3DNAARModel01best-3DNAAR-14-5-5	240	11.3
3DNAARModel01best-3DNAAR-17-5-5	240	11.3
3DNAARModel01best-3DNAAR-26-5-16	216	11.2
3DNAARModel01best-3DNAAR-17-5-16	212	10.9
3DNAARModel01best-3DNAAR-14-11-16	230	10.8
3DNAARModel01best-3DNAAR-5-11-16	230	10.8
3DNAARModel01best-3DNAAR-17-10-16	216	10.8
3DNAARModel01best-3DNAAR-14-10-16	216	10.7
3DNAARModel01best-3DNAAR-5-10-16	215	10.7
3DNAARModel01best-3DNAAR-26-10-16	208	10.3
3DNAARModel01best-3DNAAR-14-5-16	209	10.2
3DNAARModel01best-3DNAAR-26-10-5	200	10.1

⑭ After the 4MRNA is completed, further refinement is carried out using *Phenix Refine*, and the structure is finally determined.



◆ **If “No. of models” = 3**

Programming Code Generator for 4MRNA

Click here for the instructions.

Browser	Microsoft Edge
Default download directory of selected browser	C:\Users\anshi\Downloads
Working directory	F:\2TRA\4MRNA
Path and name of reflection file used in 4MRNA	F:\2TRA\Phenix\reflections.mtz
No. of models	3

Model	Type	Sequence of one strand of duplex (5'→3')	MW	No. in AU
Model 01	A-form RNA	UCC GUG ACG GGG	7638.04	1
Model 02	A-form RNA	GUU C	2449.7	1
Model 03	A-form RNA	GGC GC	3130.16	1

Generate codes

01.Python-CreatingModels-1st.py	Show this code	Download this code
02.Linux-MR-3model-Model01-1st.txt	Show this code	Download this code
03.Python-CreatingModels-2nd.py	Show this code	Download this code
04.Linux-MR-3model-Model01-2nd.txt	Show this code	Download this code
05.Linux-MR-3model-Model02-1st.txt	Show this code	Download this code
06.Python-CreatingModels-2nd.py	Show this code	Download this code
07.Linux-MR-3model-Model02-2nd.txt	Show this code	Download this code
08.Linux-MR-3model-Model03-1st.txt	Show this code	Download this code
09.Python-CreatingModels-2nd.py	Show this code	Download this code
10.Linux-MR-3model-Model03-2nd.txt	Show this code	Download this code

Download all codes

■ You can execute the codes in the same manner as when “No. of models” = 1 or 2. For reference, only the procedure is outlined below.

① Run “**01.Python-CreatingModels-1st.py**”. After execution, the first search model will be created and downloaded into 4MRNA > Model01-1, the second into 4MRNA > Model02-1, and the third into 4MRNA > Model03-1.

② Run “**02.Linux-MR-3model-Model01-1st.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model01-1st.

③ Open “results.txt” in Microsoft Excel and check parameter patterns for each of the three parameters (Tilt, Roll, Twist) that show good statistics (LLG and TFZ). Empirically, LLG ≥ 100 and TFZ ≥ 7 are typical thresholds.

④ Go to the folder 4MRNA > Model01-1 > 3DNA-○○NA, find the files corresponding to the selected parameter patterns, and copy them into 4MRNA > Model01-2. Also copy the file named “bp_step.txt” into the same folder 4MRNA > Model01-2.

⑤ Run “**03.Python-CreatingModels-2nd.py**”. After execution, new models optimized by simultaneously adjusting Tilt, Roll, and Twist will be created in 4MRNA > Model01-2.

⑥ Run “**04.Linux-MR-3model-Model01-2nd.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model01-2nd.

⑦ Open "results.txt" in Microsoft Excel to check the statistics and consider the one with the best value as the solution for the first model. Then rename the corresponding PDB file to “Model01best.pdb”.

⑧ Run “**05.Linux-MR-3model-Model02-1st.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model02-1st.

⑨ Open “results.txt” in Microsoft Excel and check parameter patterns for each of the three parameters (Tilt, Roll, Twist) that show good statistics (LLG and TFZ). Empirically, LLG ≥ 100 and TFZ ≥ 7 are typical thresholds.

⑩ Go to the folder 4MRNA > Model02-1 > 3DNA-○○NA, find the files corresponding to the selected parameter patterns, and copy them into 4MRNA > Model02-2. Also copy the file named “bp_step.txt” into the same folder 4MRNA > Model02-2.

⑪ Run “**06.Python-CreatingModels-2nd.py**”. After execution, new models optimized by simultaneously adjusting Tilt, Roll, and Twist will be created in 4MRNA > Model02-2.

⑫ Run “**07.Linux-MR-3model-Model02-2nd.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model02-2nd.

⑬ Open "results.txt" in Microsoft Excel to check the statistics and consider the one with the best value as the solution for the second model. Then rename the corresponding PDB file to “Model02best.pdb”.

⑭ Run “**08.Linux-MR-3model-Model03-1st.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model03-1st.

⑮ Open “results.txt” in Microsoft Excel and check parameter patterns for each of the three parameters (Tilt, Roll, Twist) that show good statistics (LLG and TFZ). Empirically, LLG ≥ 100 and TFZ ≥ 7 are typical thresholds.

⑯ Go to the folder 4MRNA > Model03-1 > 3DNA-○○NA, find the files corresponding to the selected parameter patterns, and copy them into 4MRNA > Model03-2. Also copy the file named “bp_step.txt” into the same folder 4MRNA > Model03-2.

⑰ Run “**09.Python-CreatingModels-2nd.py**”. After execution, new models optimized by simultaneously adjusting Tilt, Roll, and Twist will be created in 4MRNA > Model03-2.

⑱ Run “**10.Linux-MR-3model-Model03-2nd.txt**”. After execution, a file named “results.txt” will be created in the folder 4MRNA > MR-Model03-2nd.

⑲ Open "results.txt" in Microsoft Excel to check the statistics and consider the one with the best value as the solution for the third model. In addition, visually validate the fit between the structure and the electron density map using a program like *Coot*.

⑳ After the 4MRNA is completed, further refinement is carried out using *Phenix Refine*, and the structure is finally determined.